2. Introduction to Modern Civil Air Transport



2. Introduction to Modern Civil Air Transport

2.1 Facts and Figures

2.2 Market Outlook and Fleet Development

2.3 Future Challenges in Aviation

2.4 Major Players

2.5 Rules and Regulations

2.6 Safety

2. Introduction to Modern Civil Air Transport

2.1 Facts and Figures



Transport Performance in Aviation

In commercial aviation, the **passenger seat** is the production unit:

a) Available seats (AS):

$$AS = \sum_{n} (aircraft_i \cdot available\ seats_i \cdot flights_i)$$

usually given per year

b) Available seat kilometers (ASK):

$$ASK = \sum_{n} (aircraft_i \cdot available\ seats_i \cdot flights_i \cdot distance_i)$$

$$ASK = AS \cdot \overline{distance}$$

Transport Performance in Aviation

In commercial aviation, the **sold seat** is the sales unit:

a) **Number of passengers (PAX):**

$$PAX = \sum_{n} (aircraft_i \cdot passengers_i \cdot flights_i)$$

b) **Revenue passenger kilometers (RPK):**

$$RPK = \sum_{n} (aircraft_i \cdot passengers_i \cdot flights_i \cdot distance_i)$$

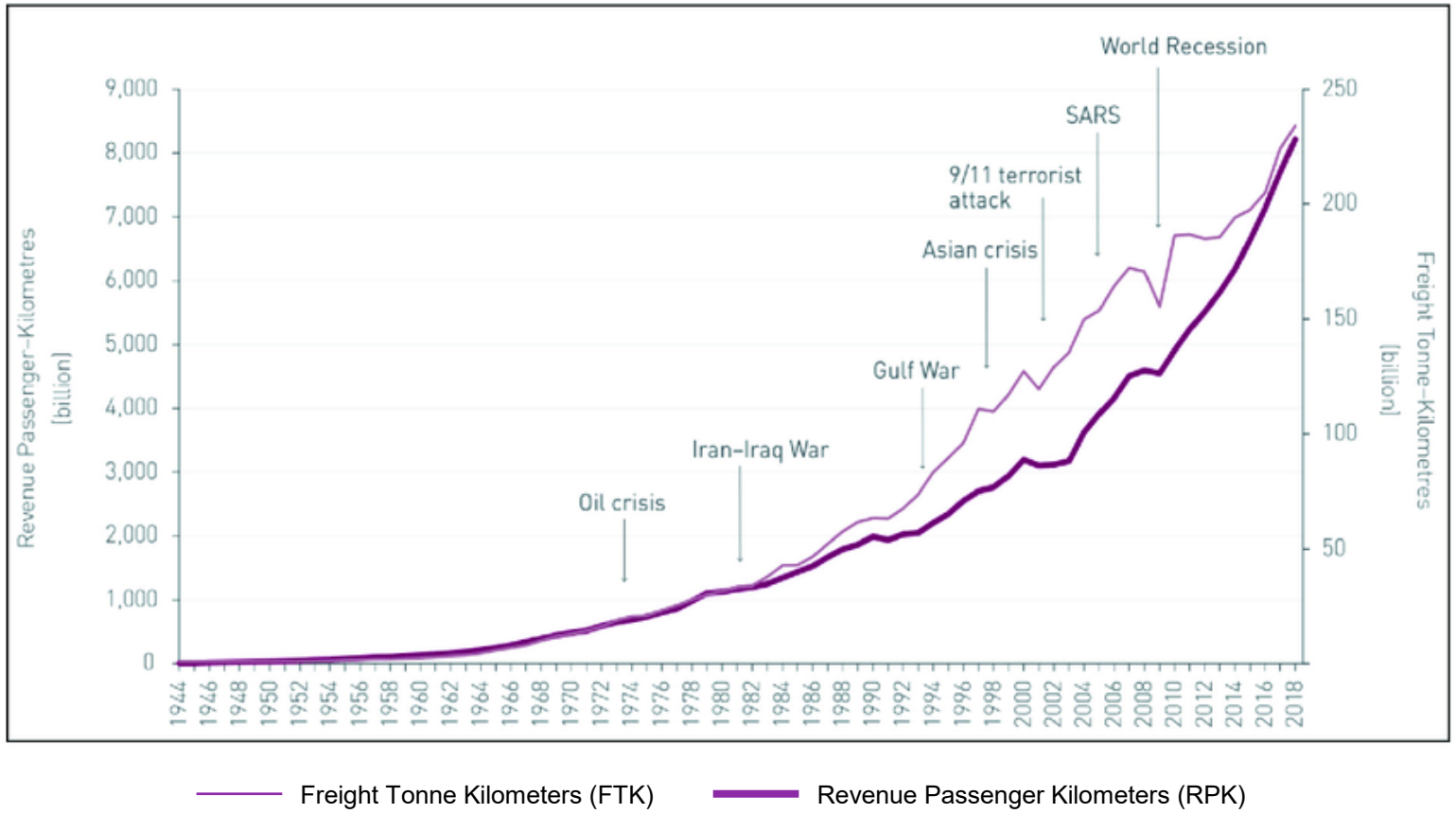
c) **Passenger load factor (PLF)**

$$PLF = \frac{RPK}{ASK} = \frac{Seats\ sold}{Seats\ available}$$

For air **cargo** calculations, the **freight ton** is the sales unit. Thus, the indices **ATK** or **ACTK** and **RTK / FTK** or **CTK** are used.

Facts and Figures – Worldwide: Air Traffic – well before COVID-19

RPK and FTK development

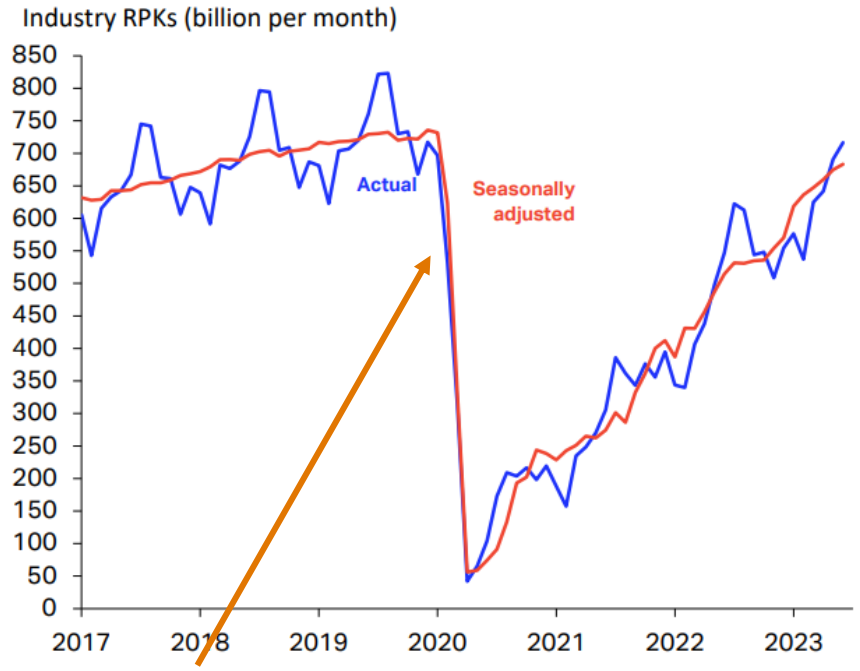


- Airbus currently expects a long term passenger traffic growth of 3.6%
- Boeing expects 4.2% annually

Source: Kiraci (2019): Financial Risk & Financial Performance: The Impact of the Global Financial Crisis on Airlines

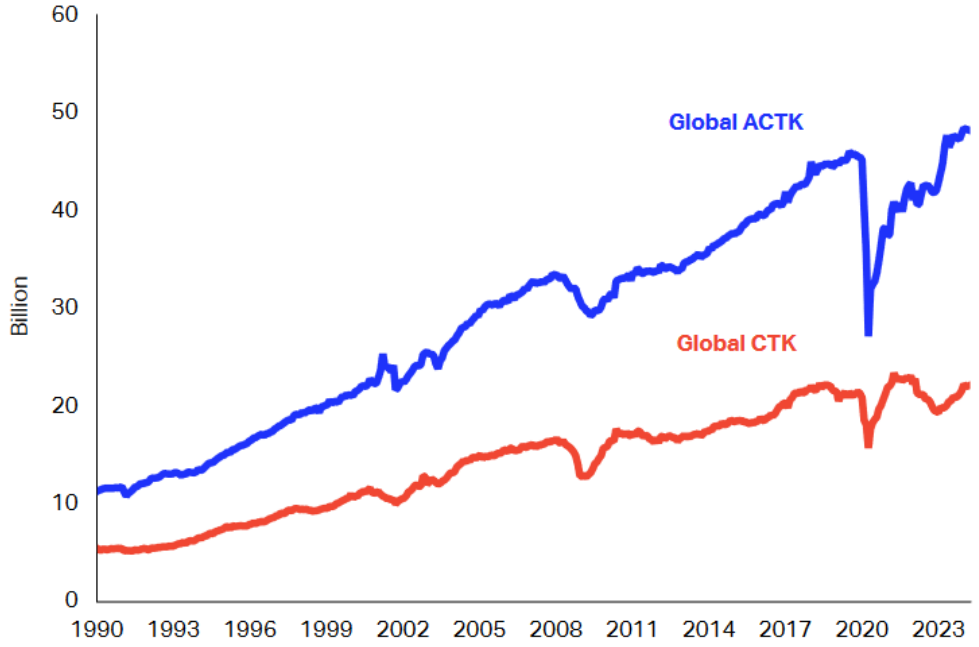
Facts and Figures – Worldwide: Air Traffic – with COVID-19

RPK



Black/Grey Swan Event:
Comes by surprise, has major impact

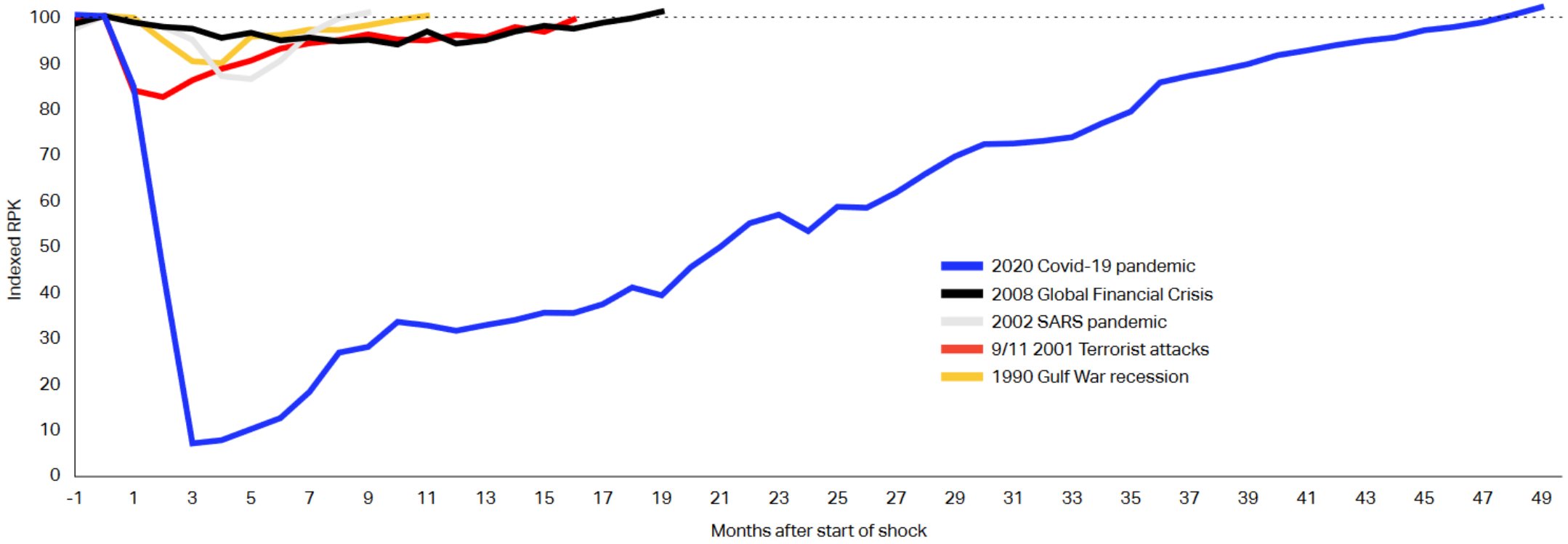
FTK/CTK development – seasonally adjusted



Source: IATA Air Passenger Market Analysis June 2023, Global Outlook for Air Transport June 2024

Facts and Figures – Worldwide: Air Traffic – with COVID-19

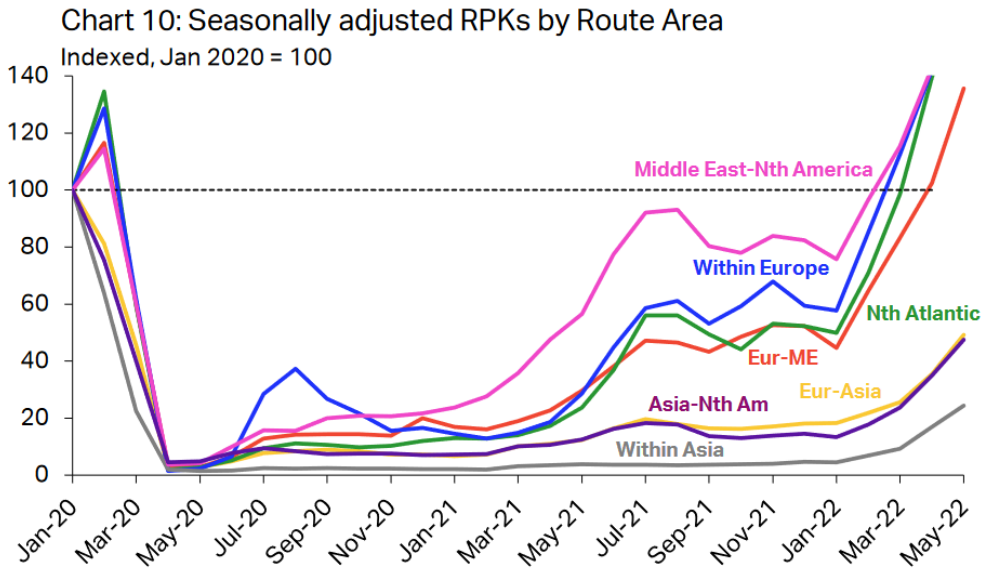
Historic recovery of industry-wide RPK after global shocks

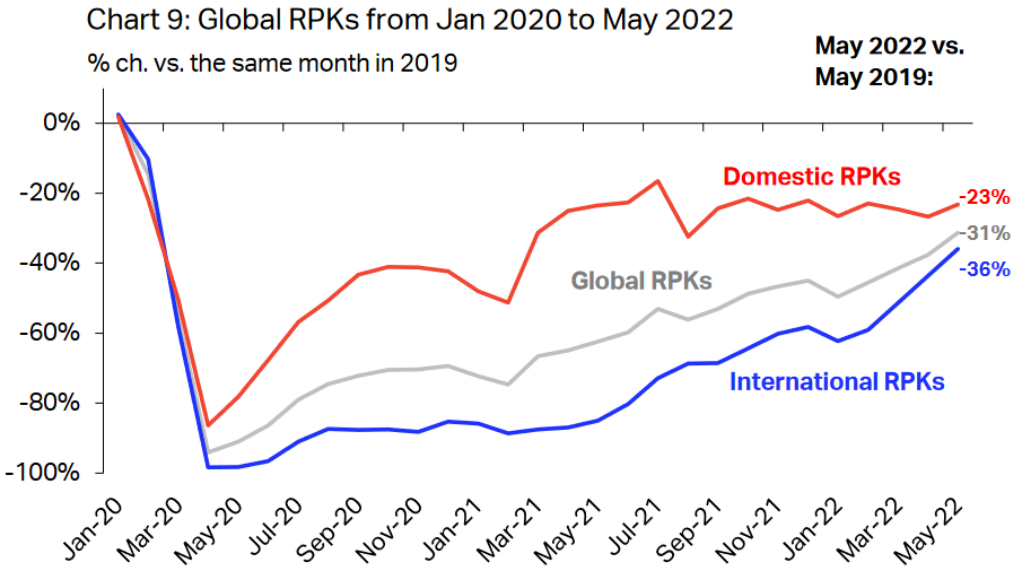


Source: Global Outlook for Air Transport June 2024

Facts and Figures – Worldwide: Air Traffic – with COVID-19

Domestic/International RPKs / Seasonally adjusted RPKs by Route Area

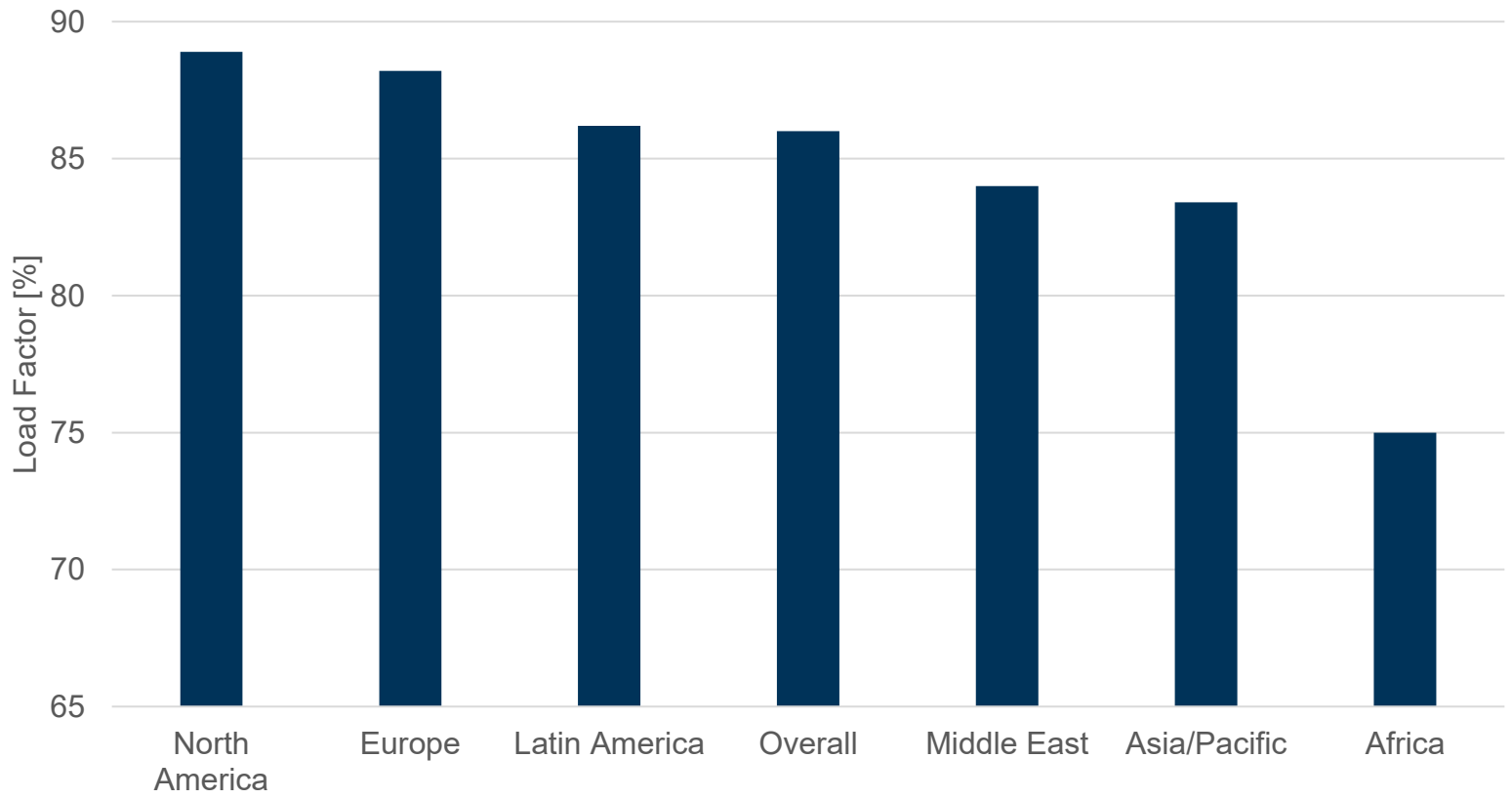




Source: IATA Quarterly Air Transport Chartbook Q2 2022

Facts and Figures – Worldwide: Air Traffic - Regional Differences

Load factors in world regions in 2024

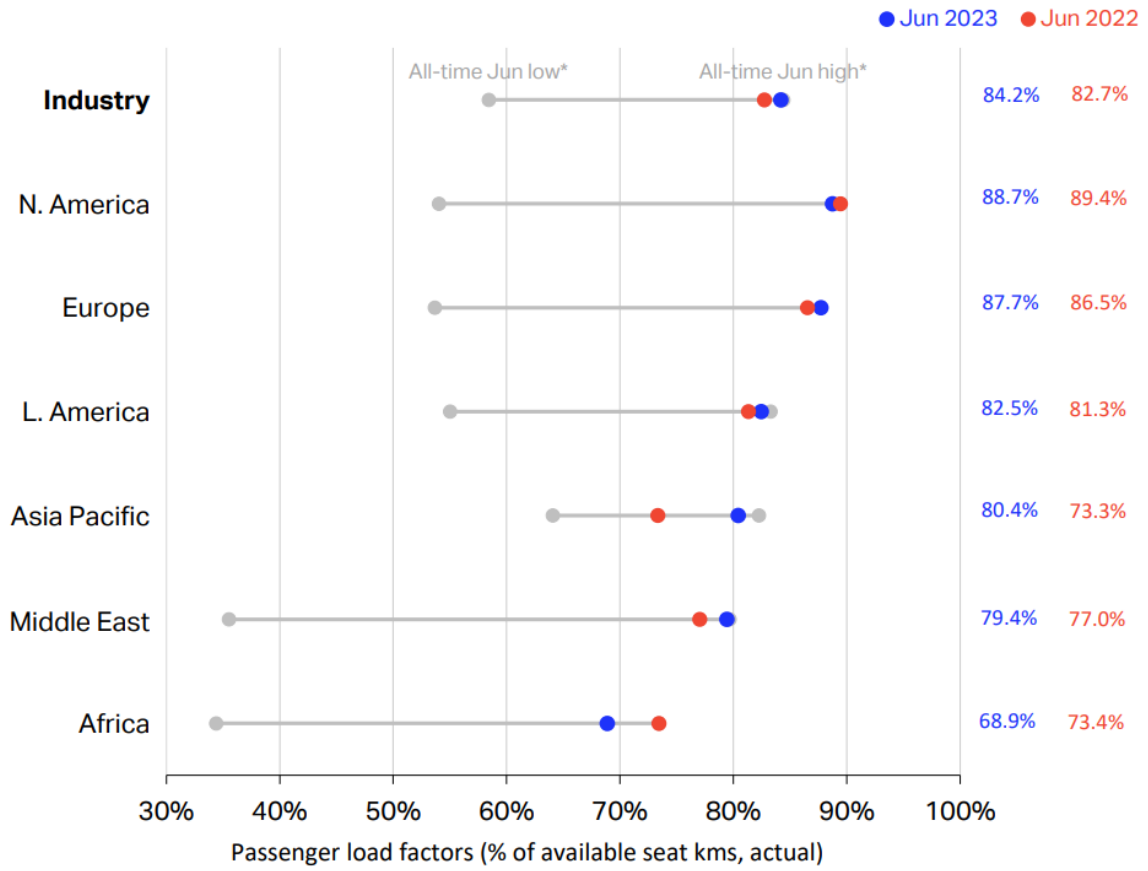


Source: IATA

Facts and Figures – Worldwide: Air Traffic – with COVID-19

Load factors in world regions in June 2023 vs. June 2022

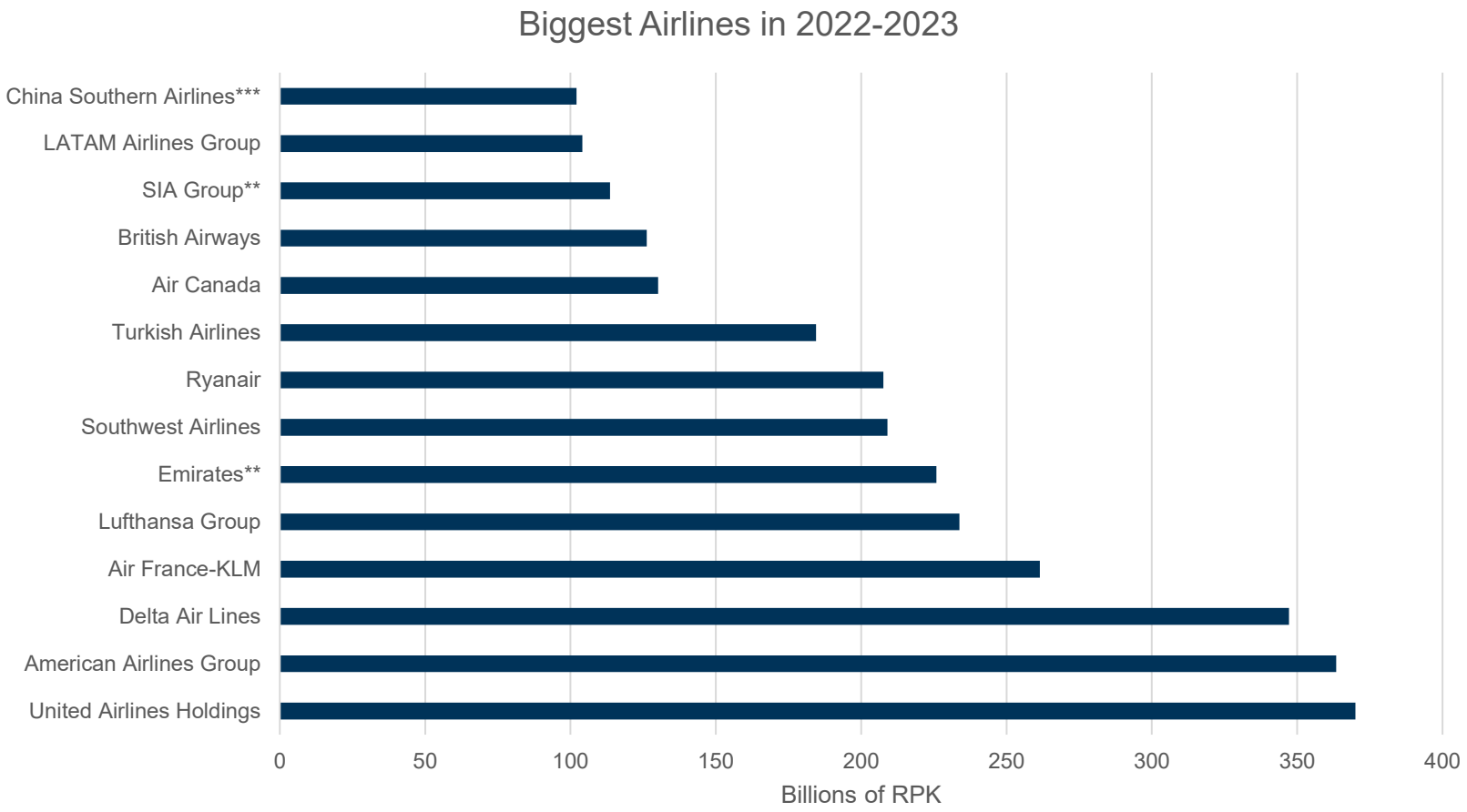
By airline region of registration



Source: IATA Sustainability and Economics, IATA Monthly Statistics, June 2023

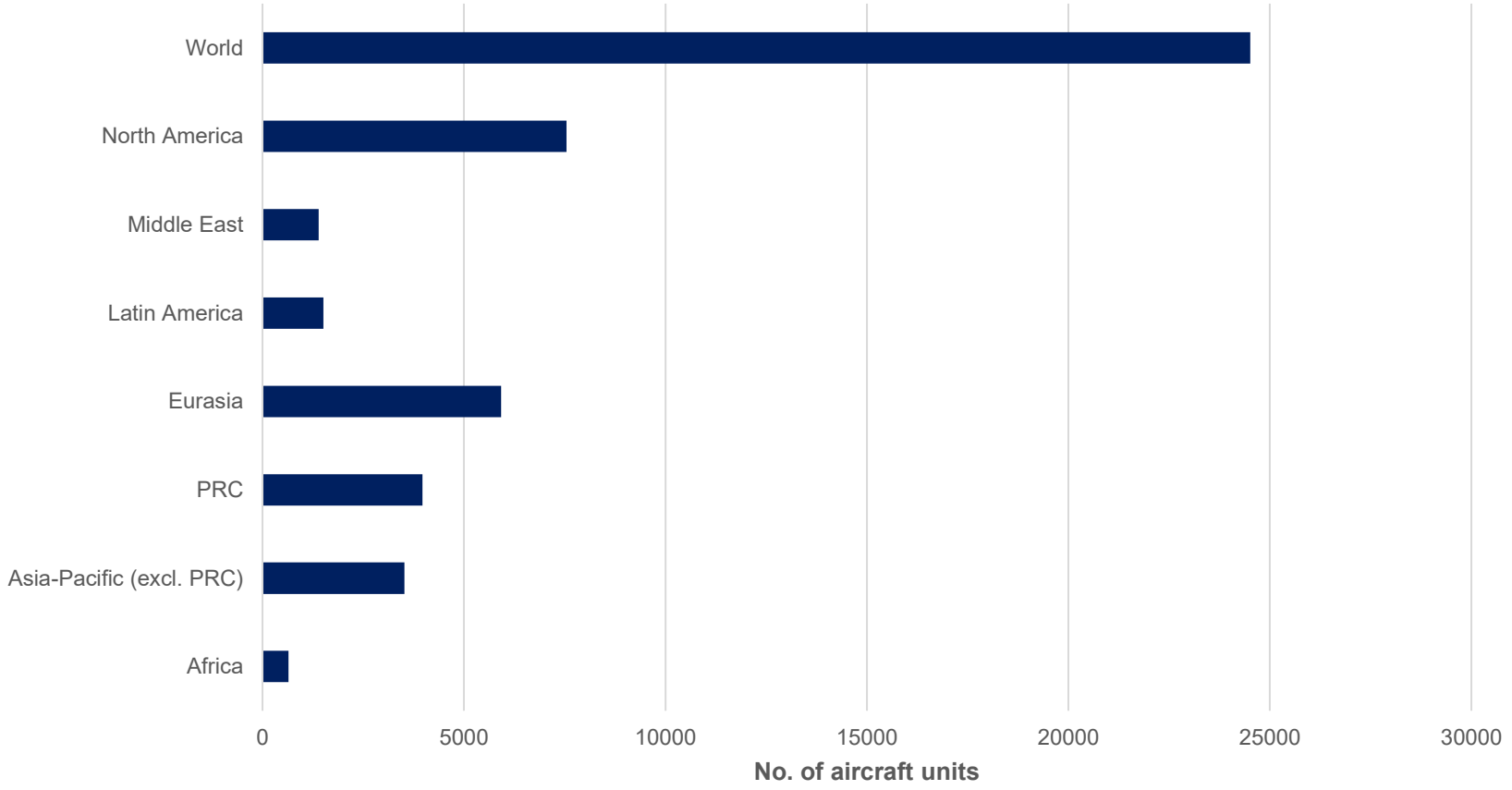
Facts and Figures – Worldwide: Airlines

Top 10 passenger airlines in 2022-2023



Facts and Figures – Worldwide: Airlines

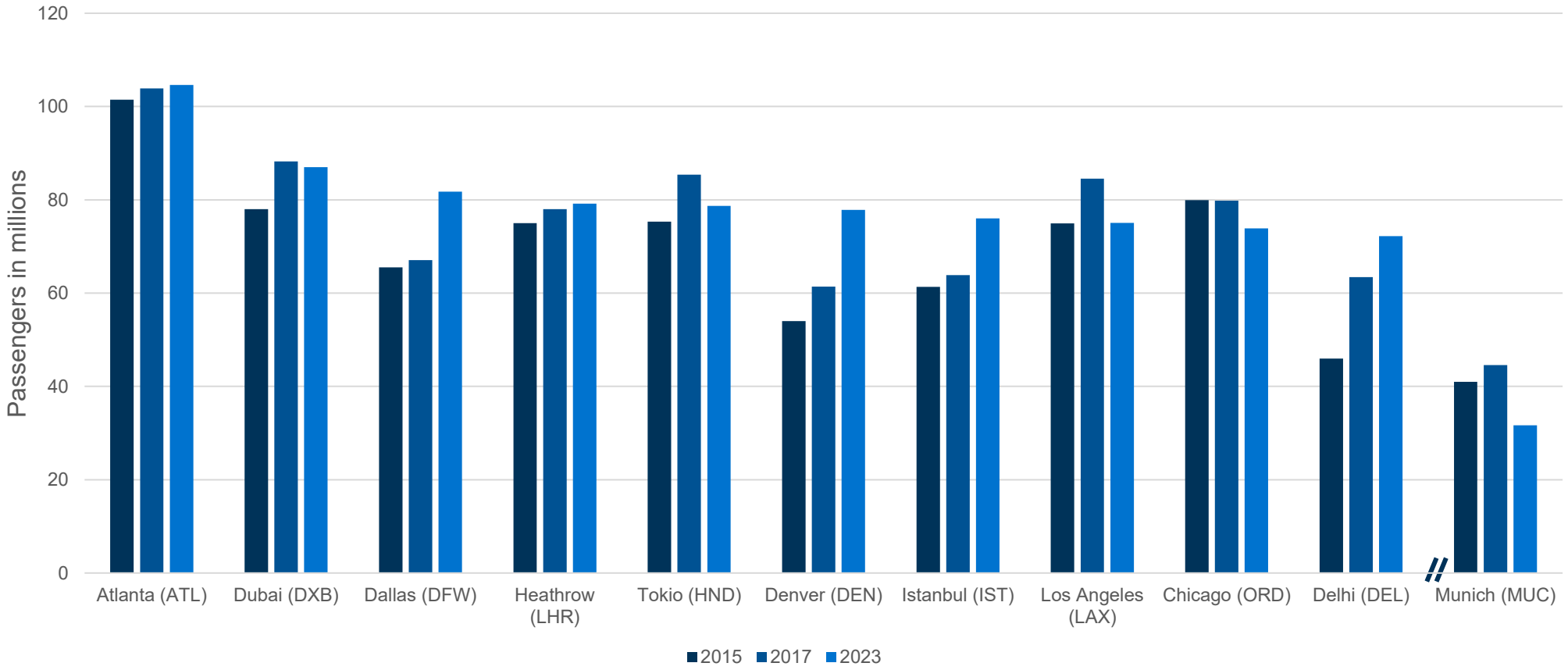
World airline fleet in 2022



Source: Boeing Commercial Market Outlook 2023

Facts and Figures – Worldwide: Airports

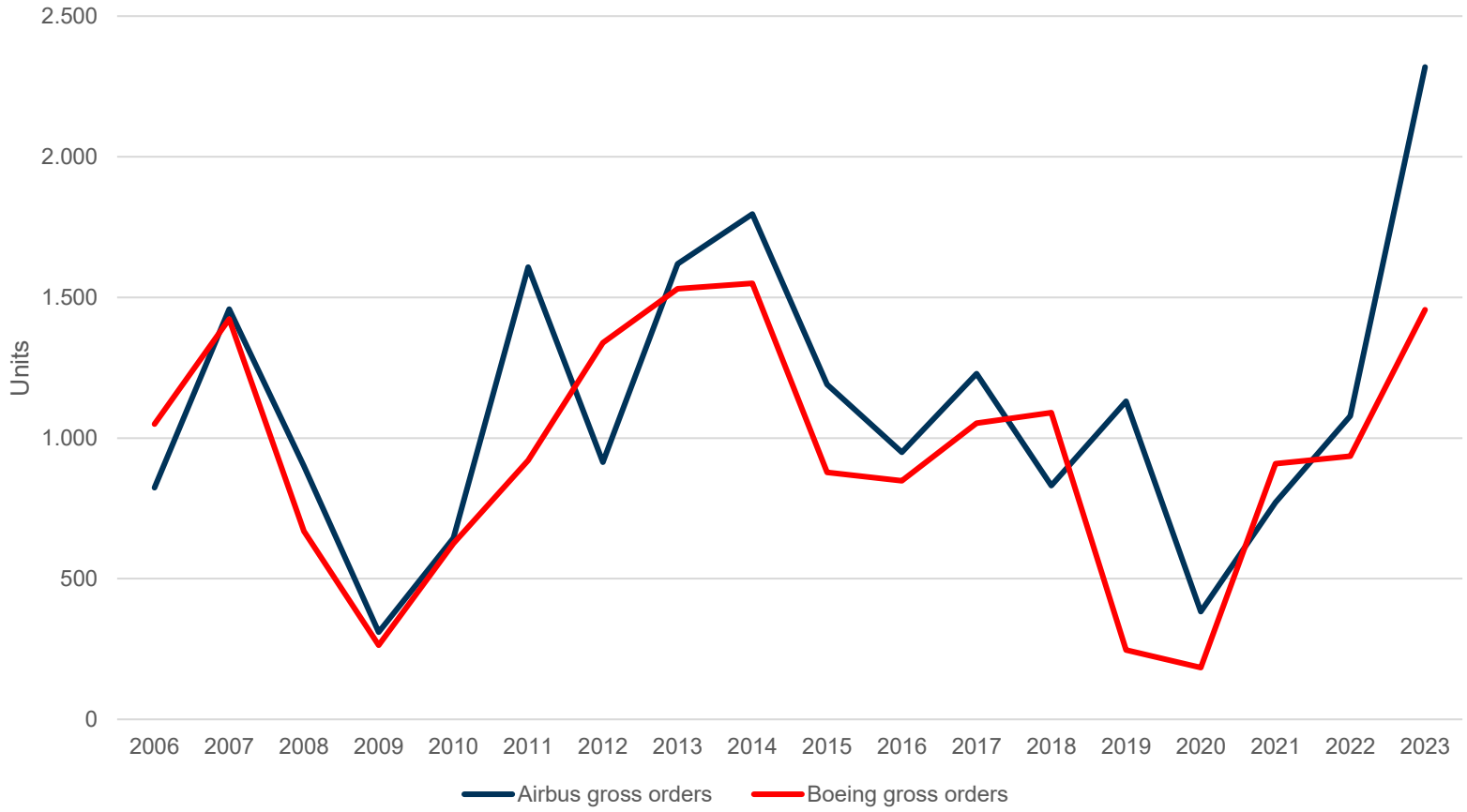
Top 10 international airports



Source: Airports Council International – Worldwide Airport Traffic Report 2022

Facts and Figures – Worldwide: OEMs

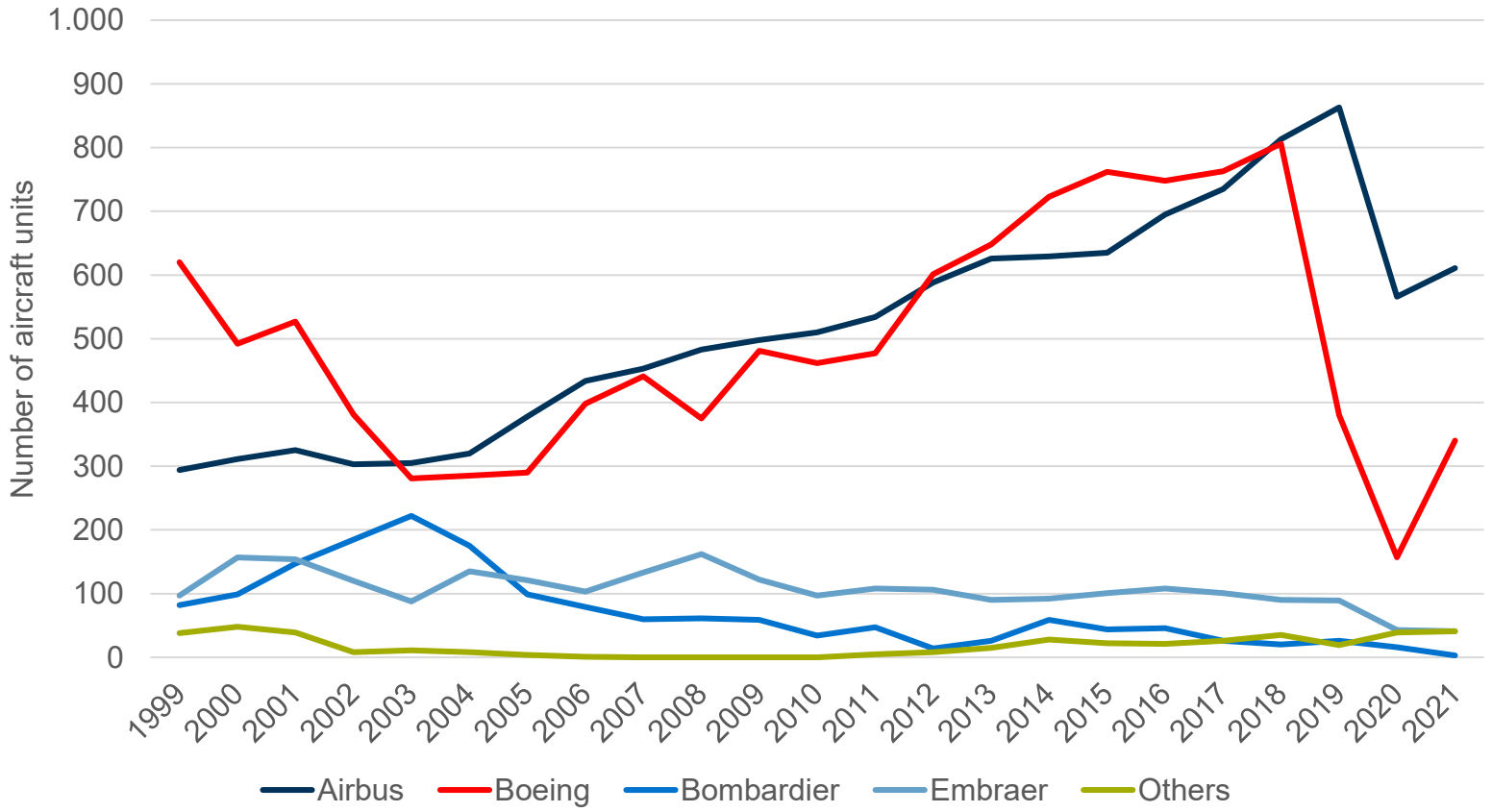
Airbus vs. Boeing: number of ordered aircraft units per year



Source: Statista 2024

Facts and Figures – Worldwide: OEMs

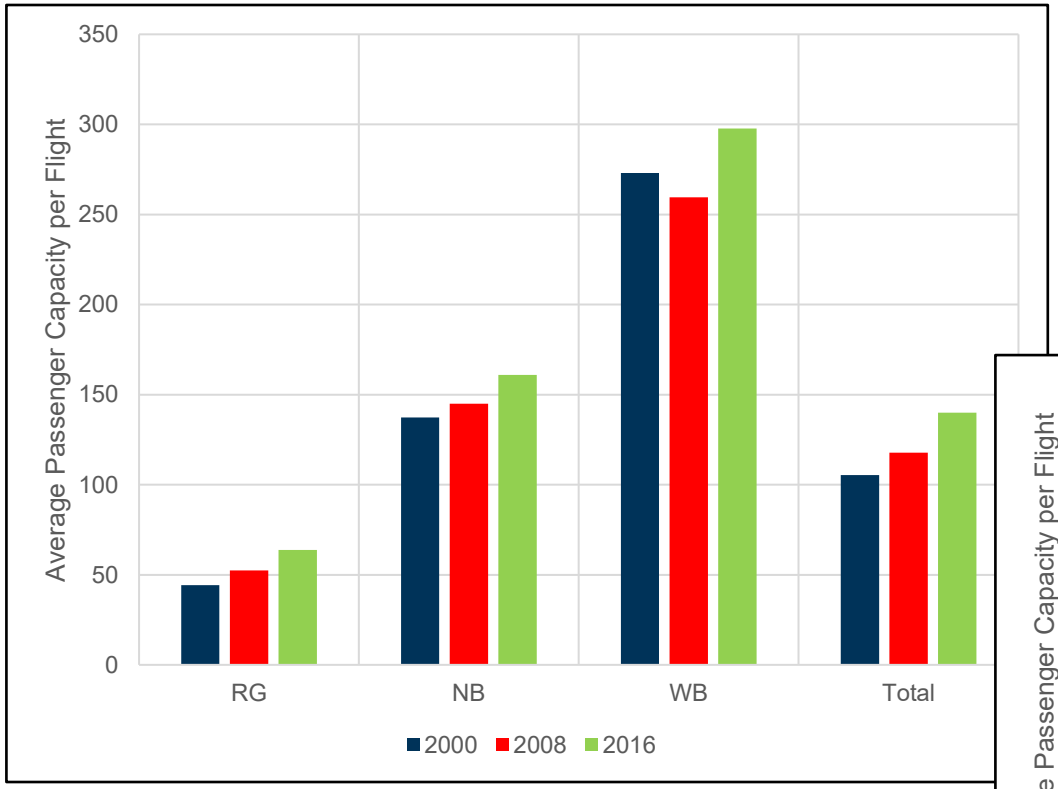
Airbus vs. Boeing: number of delivered aircraft units per year



Source: Statista

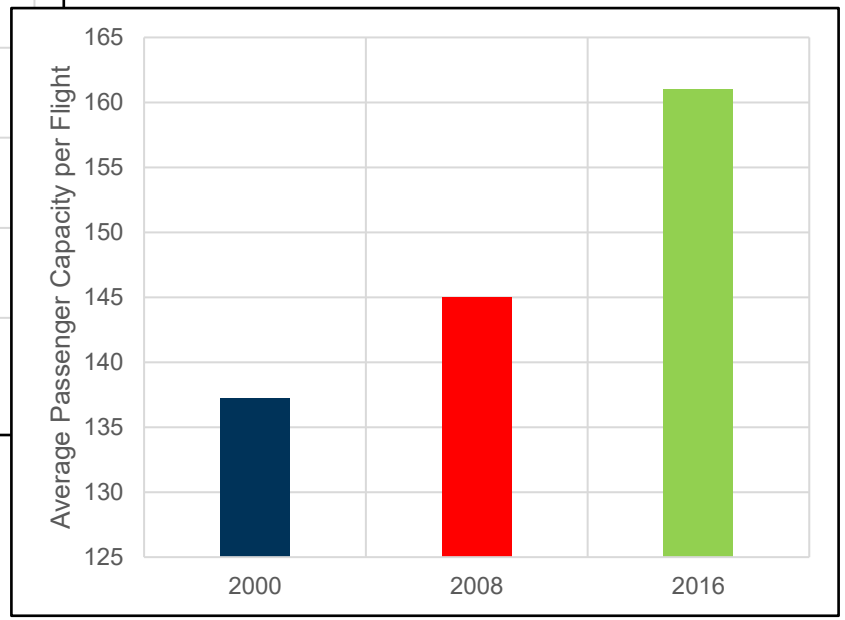
Facts and Figures – Worldwide: Aircraft Size – before COVID-19

Global development of aircraft size



Legend: RG – Regional Aircraft (Jets and Props), NB – Narrow-Body, WB – Wide-Body
Source: OAG data

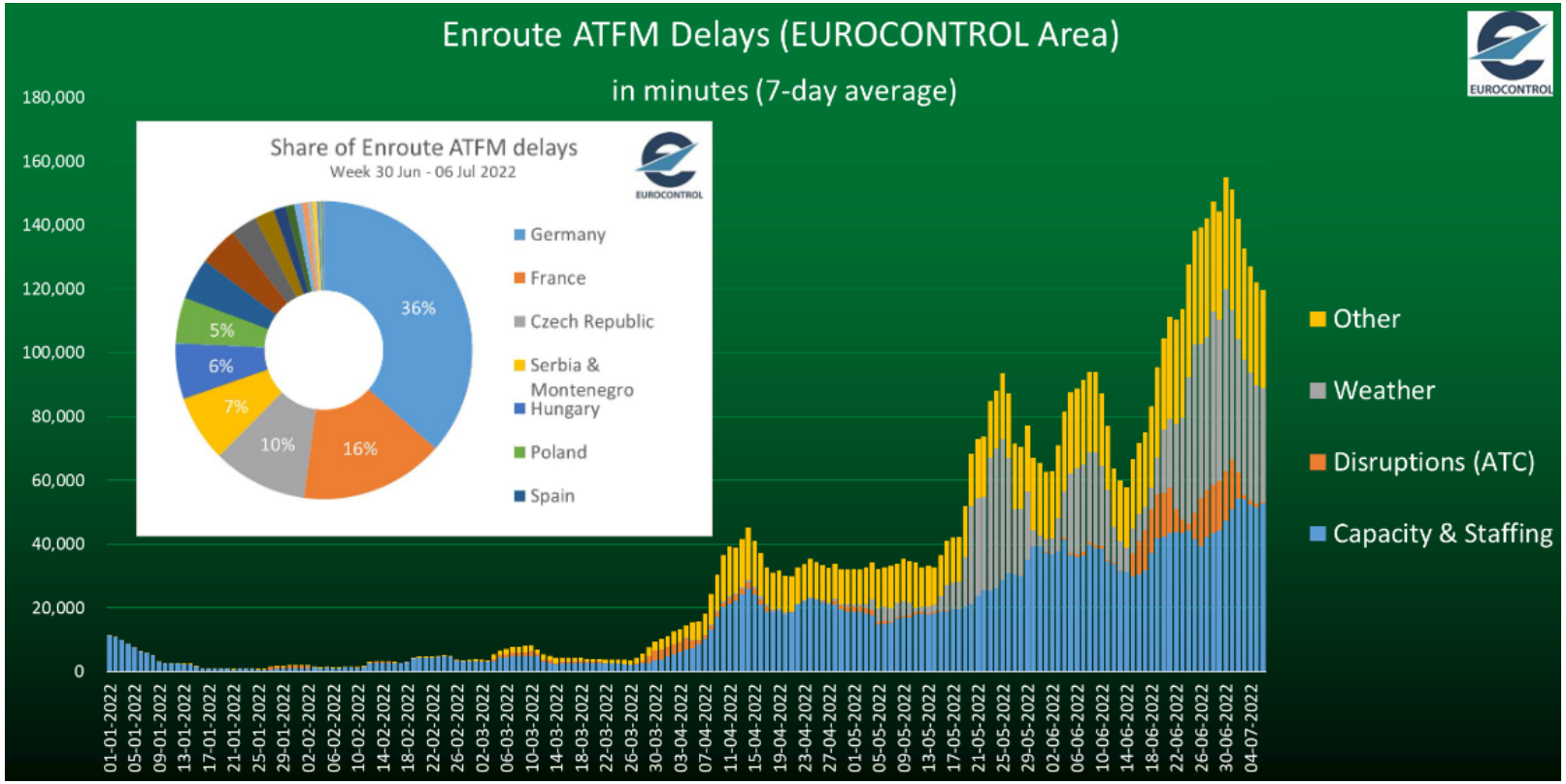
Global development of narrow-body aircraft size



Facts and Figures – Challenges after COVID 19

Congestion of air traffic system:

Too high staff layoffs during crisis, struggling airlines and airports cannot afford salary increases to attract new personnel

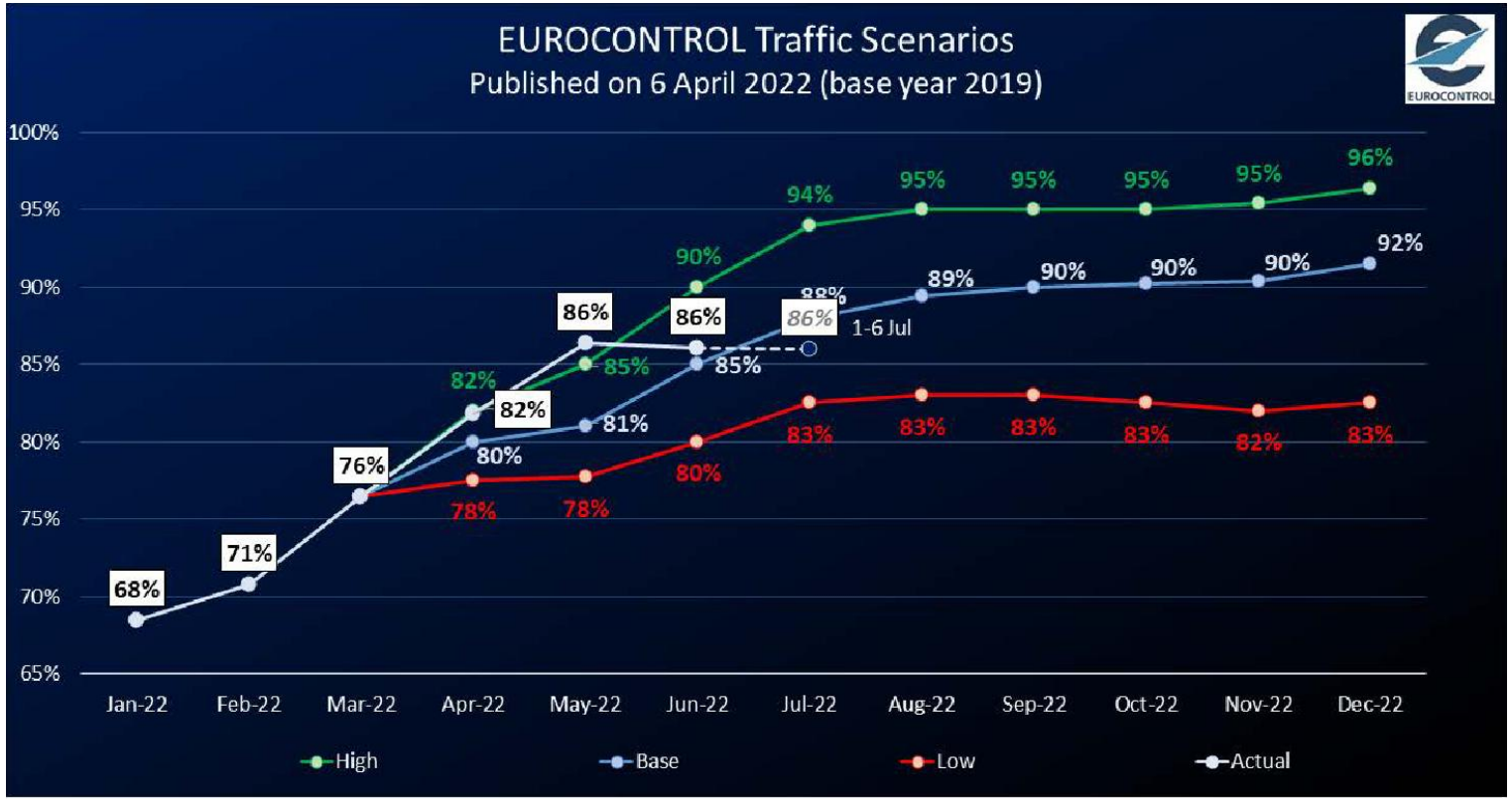


Source: EUROCONTROL

Facts and Figures – Challenges after COVID 19

Congestion of air traffic system:

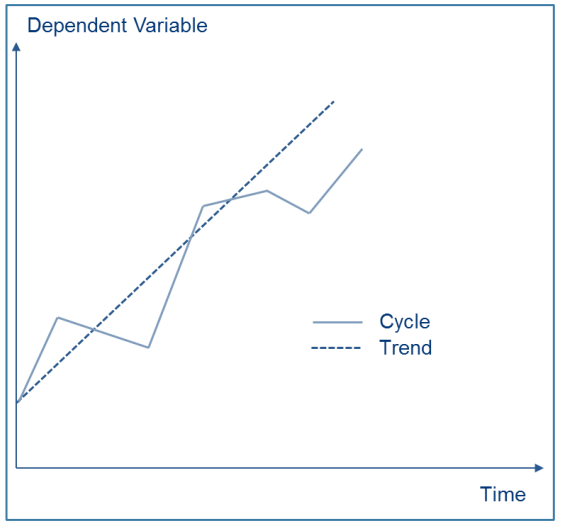
Consequence: strong demand recovery not reflected in RPK recovery anymore



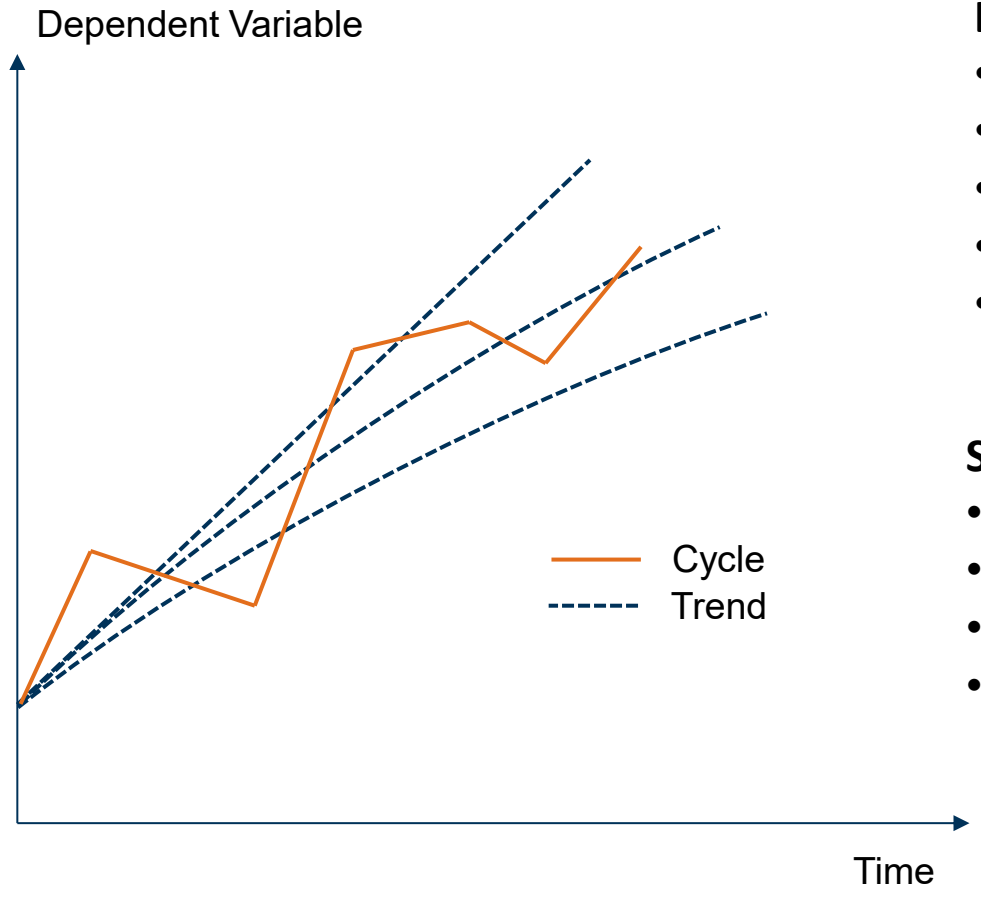
Source: EUROCONTROL

2. Introduction to Modern Civil Air Transport

2.2 Market Outlook and Fleet Development



Trends and Cycles



Long-term trends

- Investment analysis
- Assessment of new products
- Business cases
- Rules and regulations
- Industry structure

Short-term cycles

- Production rates
- Financial planning
- Subsidies
- *What-if* tests

Market Outlook Methodologies

- There are no generally applicable methods to predict market development.
- Usually, a combination of different methods is applied, most importantly
  - the ***top-down* approach (macro-approach)**
  - and the ***bottom-up* approach (micro-approach)**
- In addition, future scenarios have to be developed to capture the impact of driving factors such as
  - the environment and environmental regulations
  - Capacity constraints at airports and major air routes
  - Development of the fuel price
  - Alternative means of transport and communication

Market Outlook Methodologies – The *Top-Down* Approach

Based on the development of the past 10 – 20 years, a continuation of the prevailing trends is assumed.

The most important parameters considered here are

- **the Gross Domestic Product (GDP)**
- **Revenues of Airlines**

The world market is usually subdivided into distinct world regions (Europe, North America, Asia/Pacific, …). Air traffic within (domestic) and between (intercontinental) these regions is then analyzed.

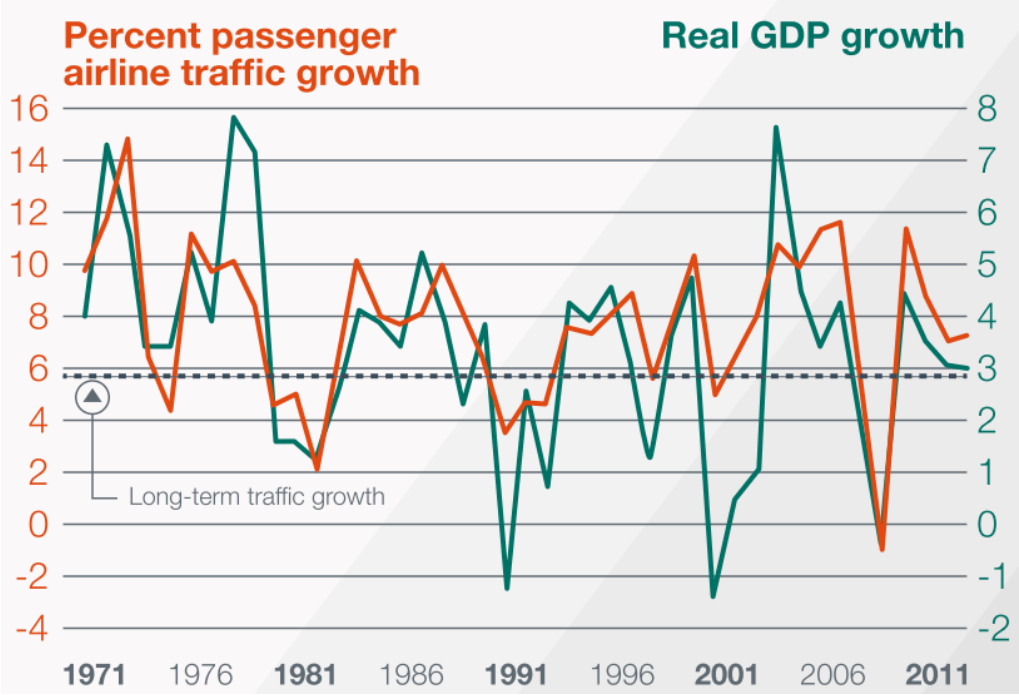
In addition, some assumptions regarding general macro-economic data are made.

The result is a long-term averaged forecast of air traffic demand (mostly 20 year into future).

Market Outlook Methodologies – The *Top-Down* Approach

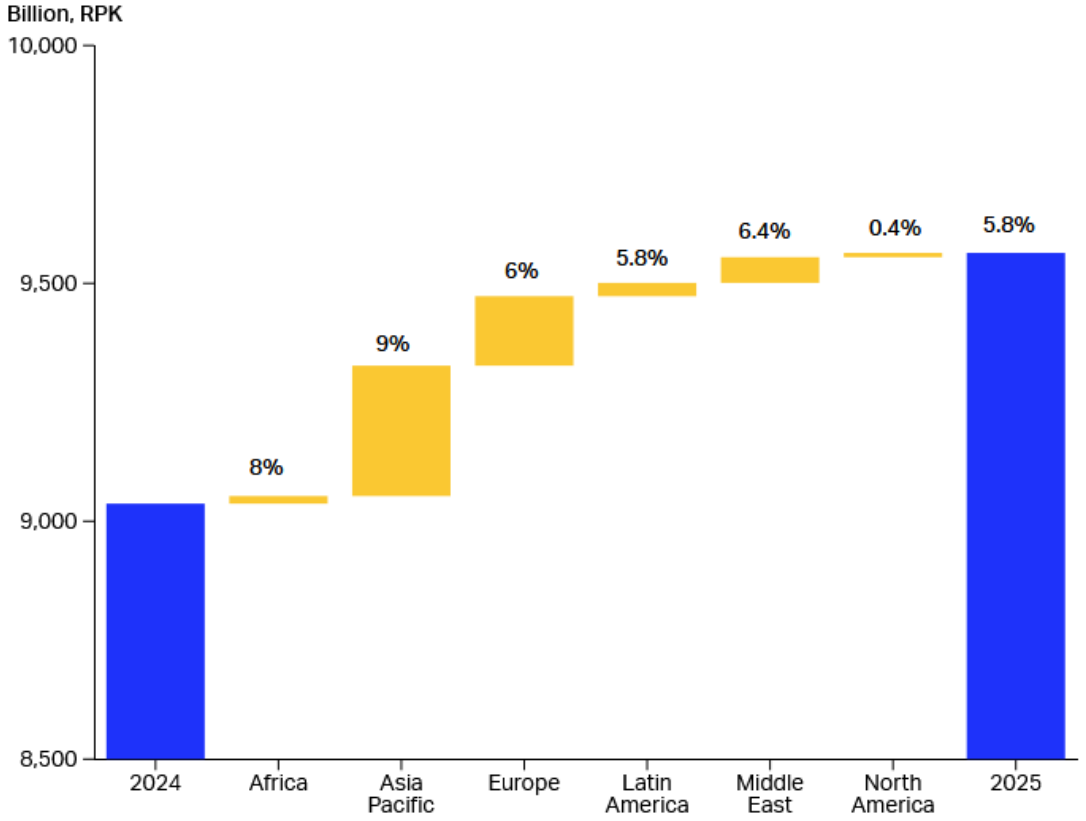
Inputs:

Passenger traffic growth VS. GDP



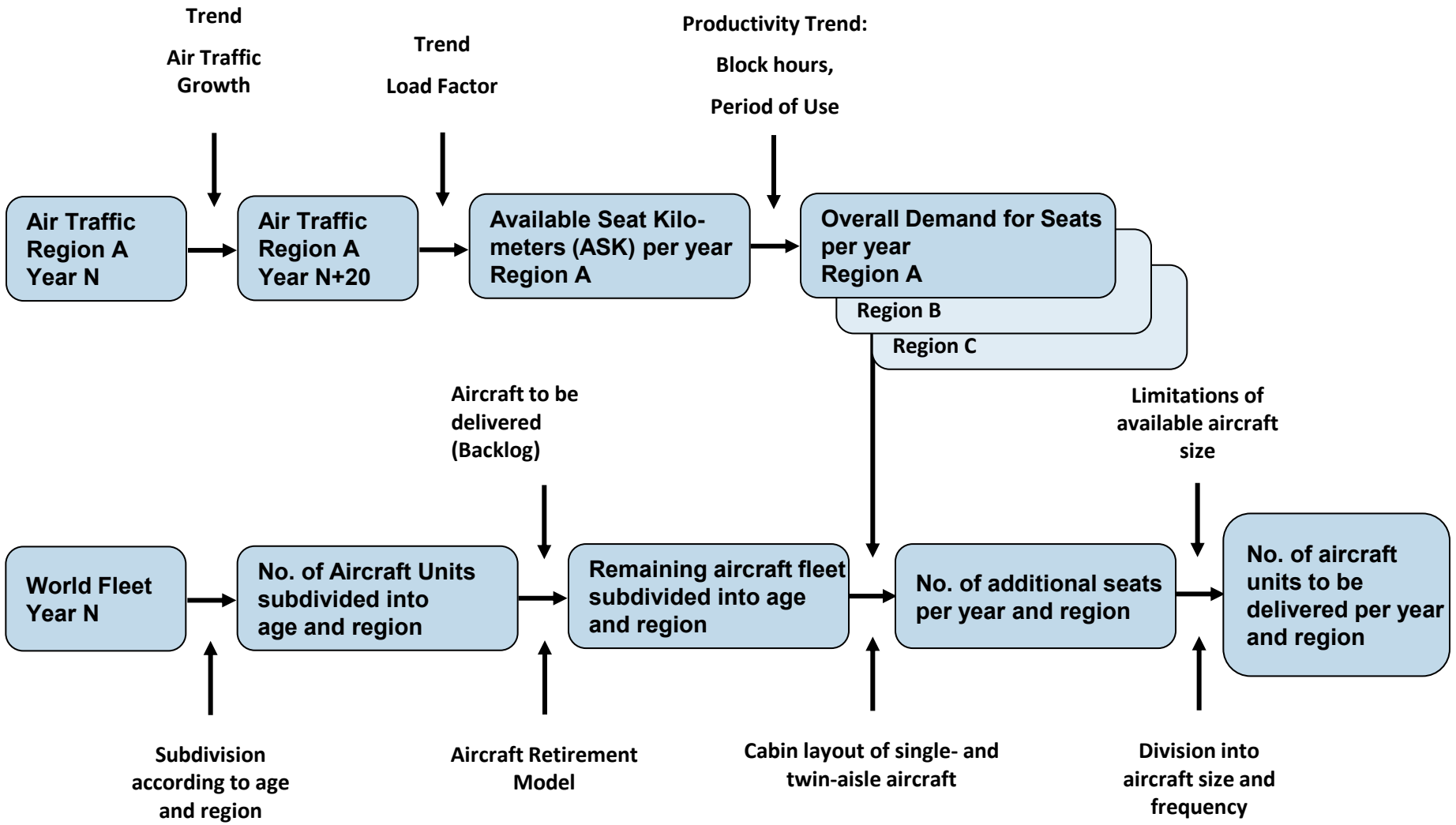
Source: Boeing CMO 2013

in different Regions



Source: IATA Sustainability and Economics June 2025

Market Outlook Methodologies – The *Top-Down* Approach

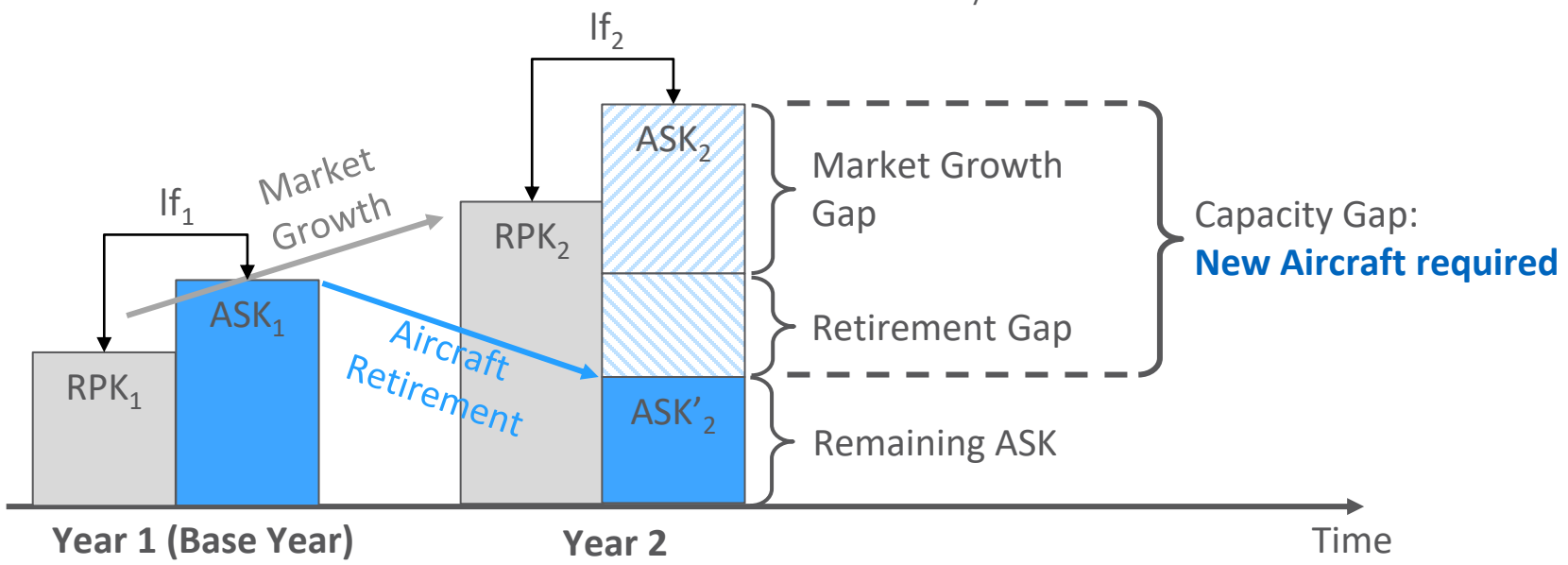


The Macro Approach to Fleet Planning

The main input data required for this approach consist of **the air transport data** and a **target average load factor**. In addition, the **retirement** of older aircraft must be considered.

Airline Transport Performance:

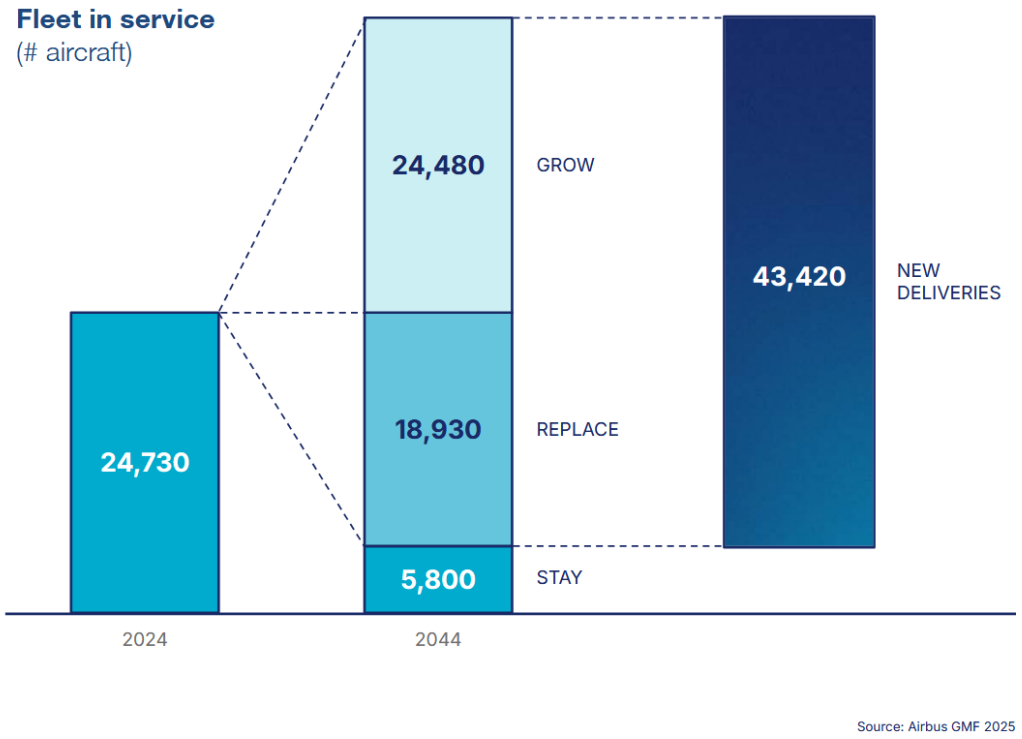
$$\mathrm{RPK} = \mathrm{LF} \cdot \mathrm{ASK} = \mathrm{LF} \cdot \sum_{\mathrm{a/c}} (\mathrm{Seats} \cdot \mathrm{Dist} \cdot \mathrm{Freq})$$



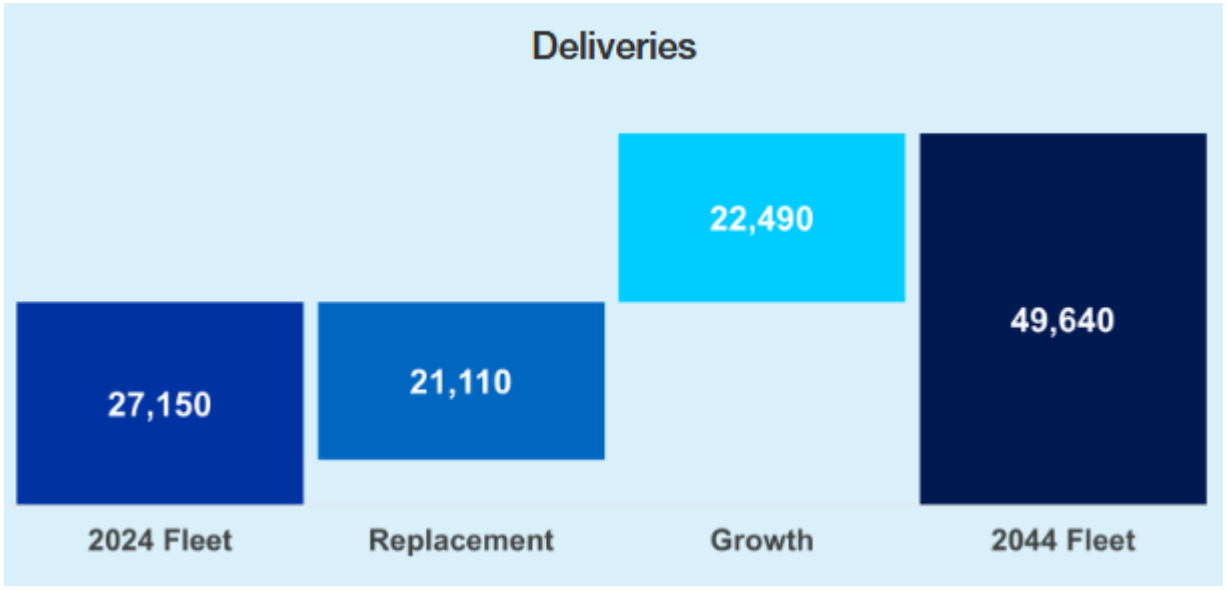
...the same applies to ATK.

Source: Belobaba (2009), Clark (2007)

The Macro Approach to Fleet Planning



Source: Airbus GMF 2025 (2024),



Boeing CMO 2025 (2024)

Aircraft Utilization

For Fleet Planning it needs to be taken into account that aircraft do not spend all the time in the air:

Utilization (UH) = Block Hours (BH) + Turn-Around Hours (TH) + Maintenance Hours (MH)

BH = Time needed to accomplish flight mission from gate to gate

BH = f (stage length, cruise speed, airport infrastructure, etc.)

TH = Time needed to de-/board the aircraft and make it ready for the next flight mission

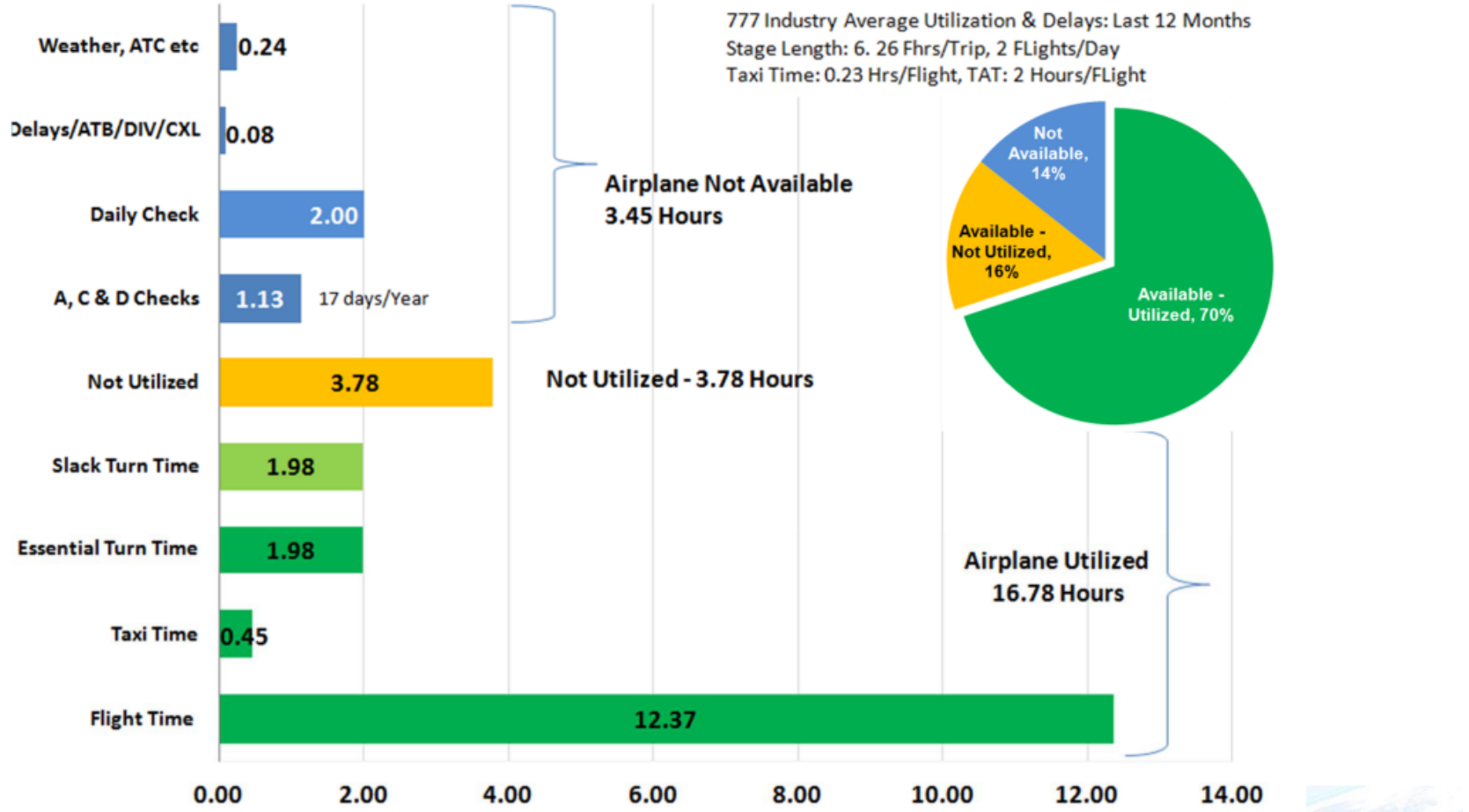
TH = f (no. of passengers, type of aircraft cabin, airport infrastructure, etc.)

MH = Time needed to maintain the aircraft

MH = f (type of aircraft, type of engines, degree of utilization, etc.)



Example: How a Boeing 777 spends a Day

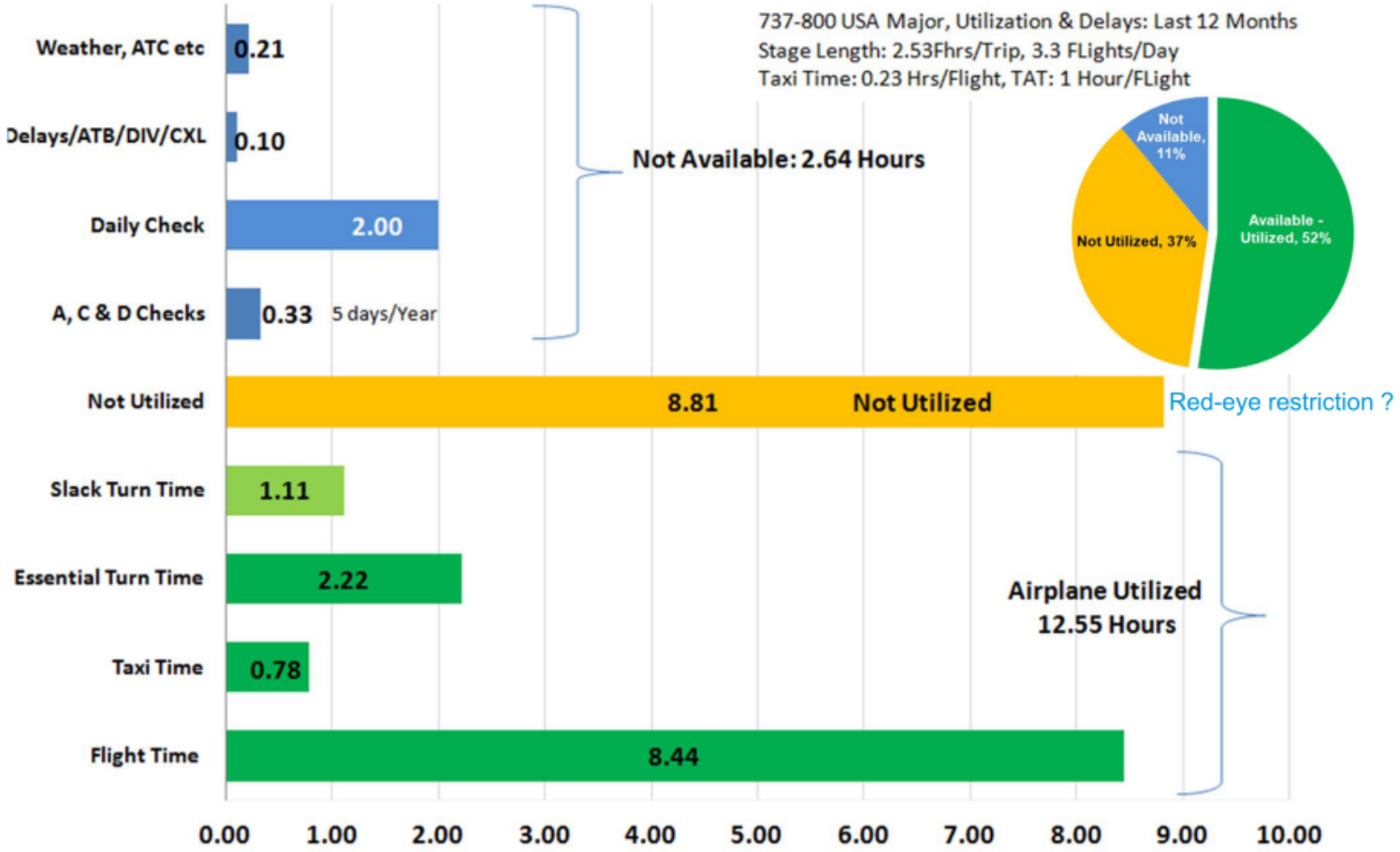


Aircraft Availability: 86%, but used only 70 %!



Source: Boeing (2013)

Example: How a 737NG spends a Day with major US Airline



Aircraft Availability: 89%, but used only 52%!



Source: Boeing (2013)

Exercise: Macro Approach / Utilization (1)





What **type of aircraft** is suitable for an airline in a long-term scenario for a **given flight schedule**?

- Airline flies from **Singapore (SIN)** to **Tokyo Narita (NRT)**
- **Distance d_{avg}: 5314 km**
- Expected $\boldsymbol{RPK_{2020}}$**: $\boldsymbol{9.5 \cdot 10^8}$ PAXkm**
- Optimal **load factor** for the airline $\boldsymbol{LF}$**: 85 %**
- purchased slots: **One inbound** and **outbound** flight a day
- 2 alternative aircraft types: Boeing 787-10 (**344 seats**) and A350-900 (**293 seats**)

$$n_{seatsperaircraft} = \frac{RPK_{2020}}{n_{flights} \cdot LF \cdot d_{avg}} = 289\ seats$$

- nearest aircraft on **RPK target: *Airbus A350-900***
- $\boldsymbol{LF = 83.6}$ % for the A359, $\boldsymbol{LF = 71.2}$ % for the B78X

Picture source: worldairlinenews.com, airlinereporter.com

Exercise: Macro Approach / Utilization (2 – Growth and Utilization)

Block time SIN-NRT BH: **7 h**
Maintenance time MH: **2 h**
Tourn-around-time at NRT TH: **1 h**
Due to maintenance regulations **2 aircraft needed**
Annual traffic growth GF 2020-2022 of **5 %** expected
same number of flights as before



What ***RPK*** **are expected with traffic growth**? Same aircraft possible?

$$RPK_{2022} = RPK_{2020} \cdot (1 + GF)^{2022-2020} = 1.047 \cdot 10^9\ PAXkm$$

New load factor LF to high for aircraft type? -> passenger spilling costs?

$$LF_{A359,2022} = \frac{RPK_{2022}}{n_{flights} \cdot n_{seatsperaircraft,A359} \cdot d_{avg}} = 92.2\ \%$$

✗ Too high

$$LF_{B78X,2022} = \frac{RPK_{2022}}{n_{flights} \cdot n_{seatsperaircraft,B78X} \cdot d_{avg}} = 78.5\ \%$$

✓ More growth possible

What is the **utilization** per aircraft and day? Can **more flights** be offered?

$$UH = \frac{n_{flights} \cdot (BH + MH + TH)}{n_{ac} \cdot 365\ d} = 10h/d \rightarrow 41.7\ \%$$

✓ More flights possible

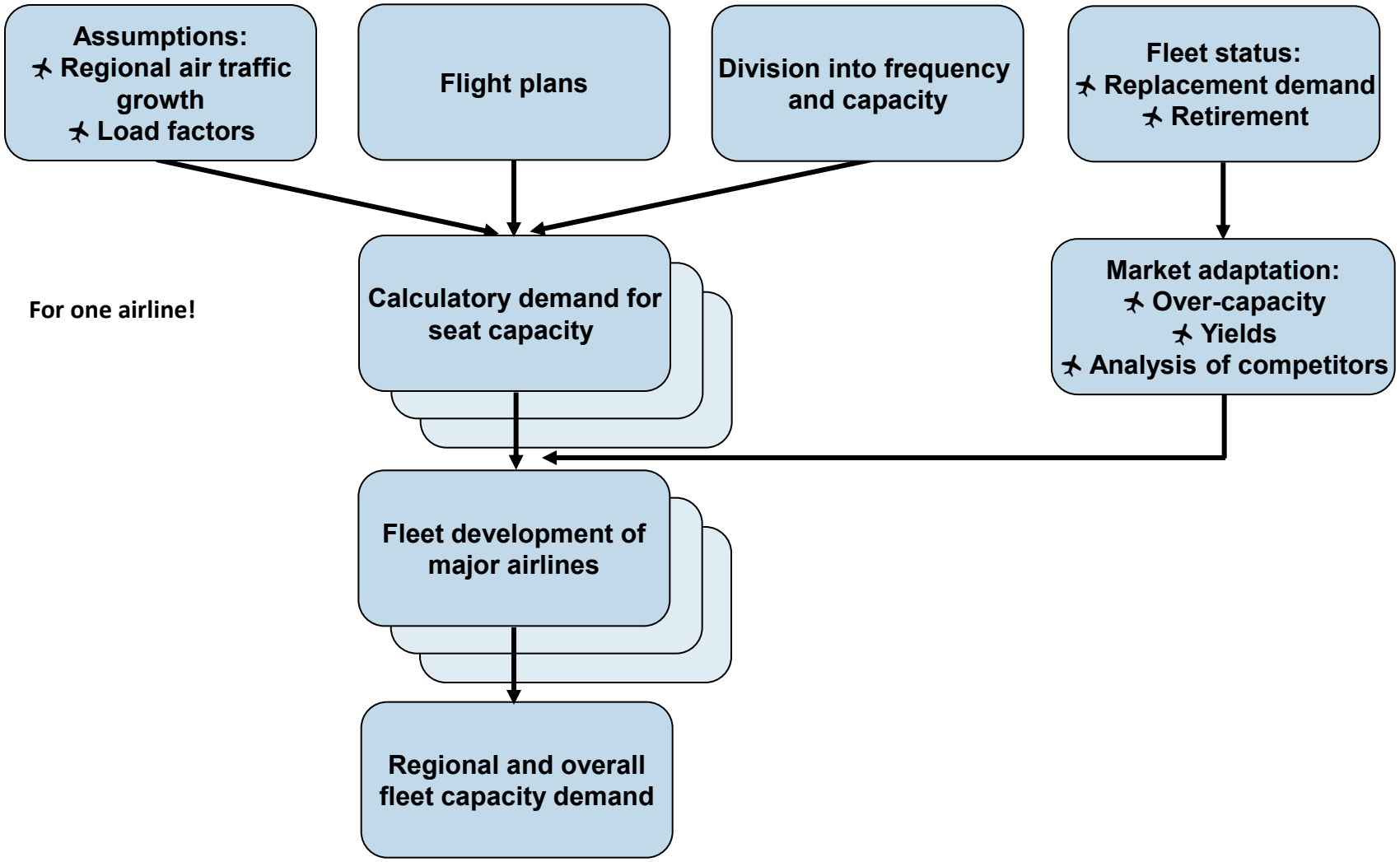
Picture source: fraport.com

Market Outlook Methodologies – The *Bottom-Up* Approach

The *bottom-up* approach

- is normally used for internal purposes within a company only
- is based on the analysis of a single airline by analyzing its fleet structure
- requires continuous updating and adaptation efforts of the data and is therefore very work-intensive
- is usually employed for purposes of production planning, evaluation, and the positioning of new products
- can only provide a limited meaning for aircraft with less than 100 seats and aircraft with more than 350 seats

Market Outlook Methodologies – The *Bottom-Up* Approach



The *Bottom-Up* Approach

- …involves a much more detailed evaluation of **specific route characteristics** and **aircraft requirements**.
- …requires much **more detailed descriptions of future scenarios** for the subnetwork or routes under consideration (e.g., traffic growth rates of specific Origin-Destination Markets)
- …generates **much more detailed outputs** that include individual aircraft tail assignments and operating statistics by route, flight, and even aircraft tail number.
- The airline's own share of total demand in each O-D market must be estimated with a **market share model**.
- Forecasts of demand and revenues for each O-D market are then allocated to each future flight in the airline's schedule using a **traffic allocation model**.

➢ The *Bottom-Up* Approach **provides the airline with a complete representation of its network and operations** under different fleet alternatives for a range of time periods into the future.

➢ The *Bottom-Up* Approach requires **highly detailed input data** featuring **high uncertainty about the future**. The Macro Approach is therefore more commonly used.

Source: Belobaba (2009), Clark (2007)

Air Traffic Market Outlook: Industry Reports

Airbus Global Market Forecast



https://www.airbus.com/sites/g/files/jlcbta136/files/2024-07/GMF%202024-2043%20Presentation_4DTS.pdf

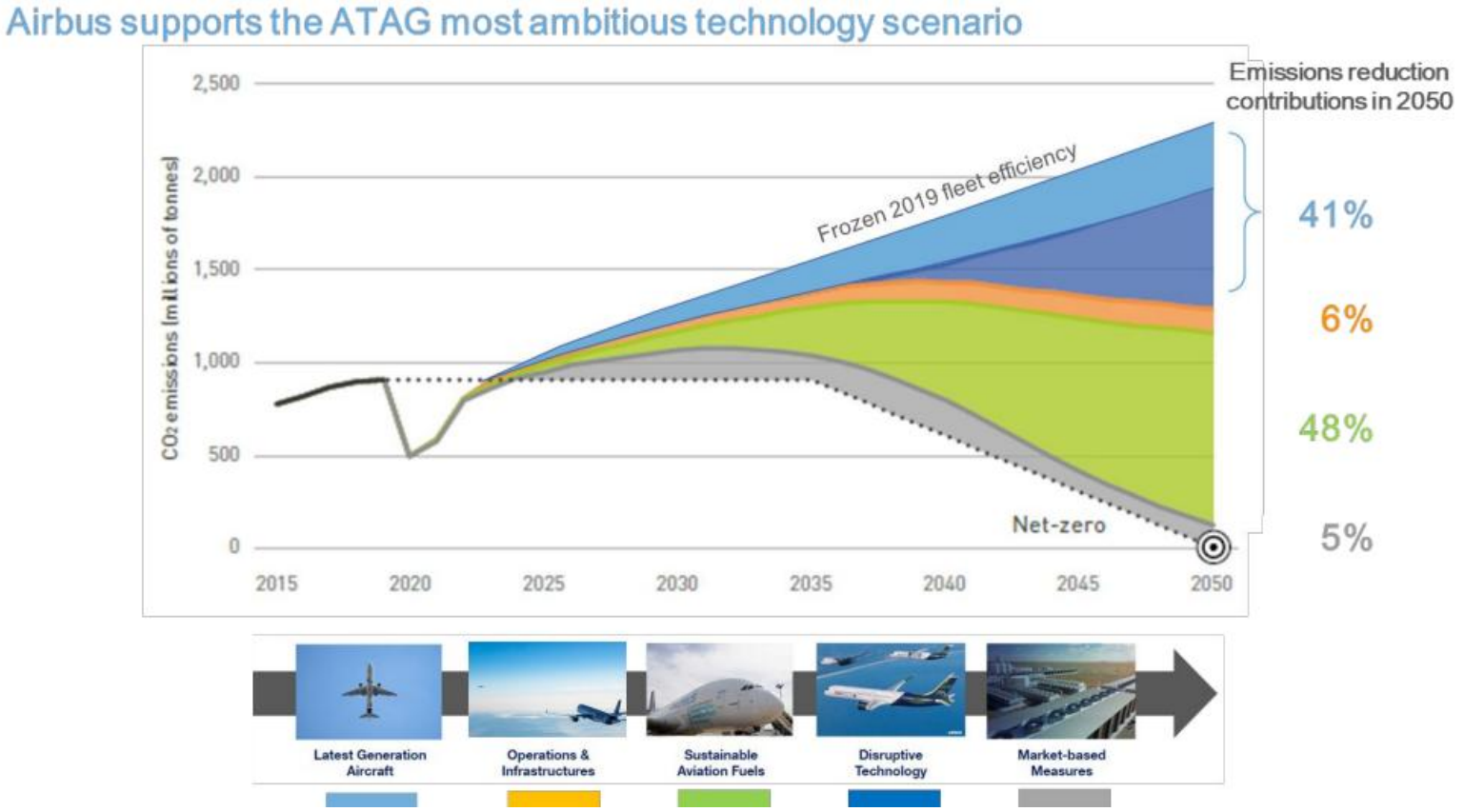
Boeing Commercial Market Outlook



https://www.boeing.com/content/dam/boeing/boeingdotcom/market/assets/downloads/2024-cmo-executive-presentation.pdf

Growth of Air Traffic & Regional Fleet Distribution

Example from Airbus:



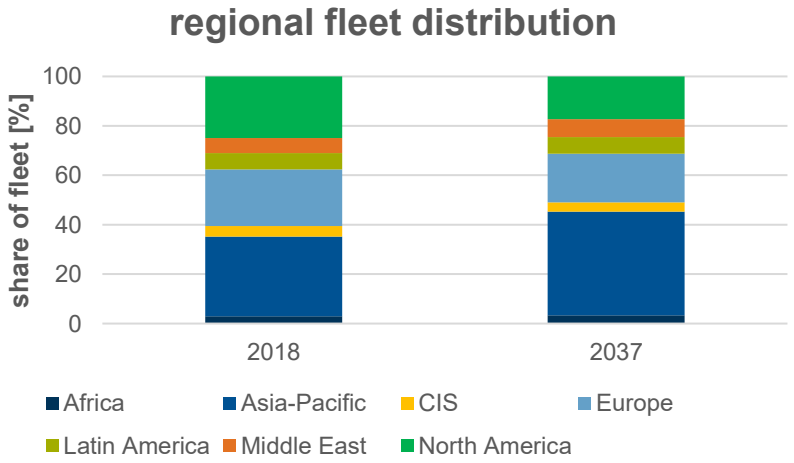
Latest Generation Aircraft

Disruptive Technology

Operations & Infrastructures

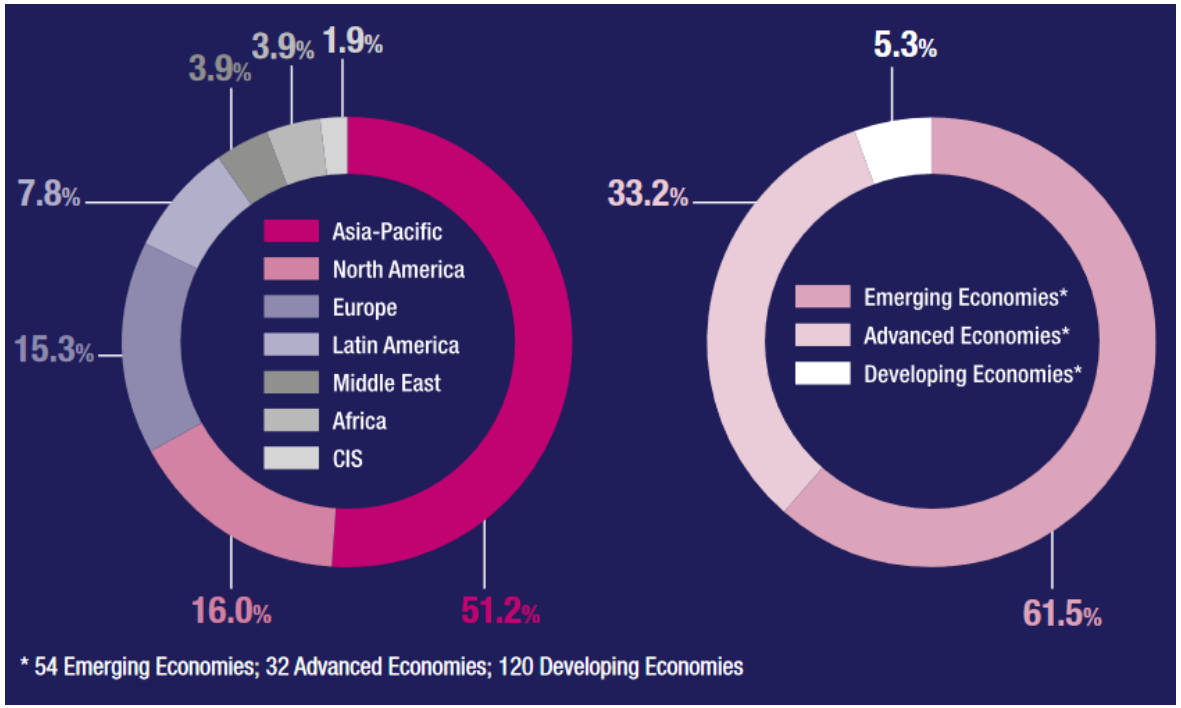
Sustainable Aviation Fuels

Carbon offsetting and capture



Different Developments in World Regions – Reasons

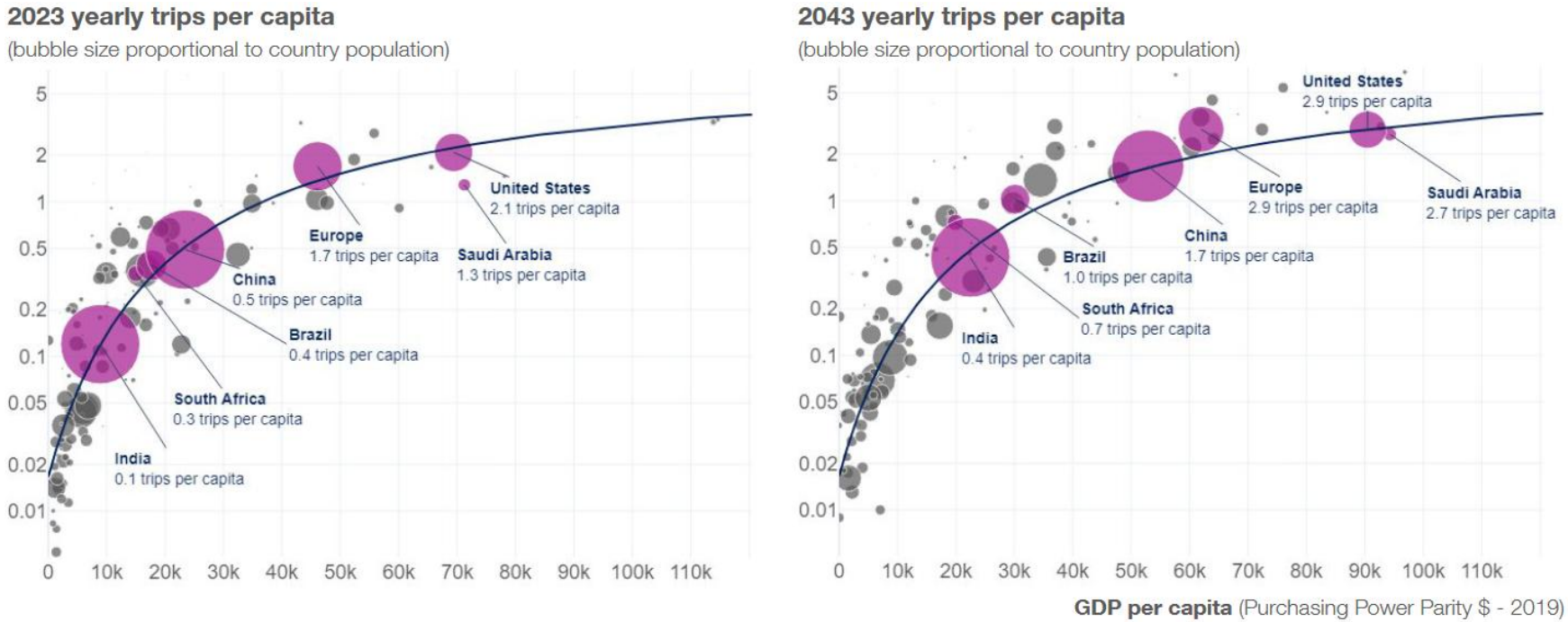
- Different degrees of economic and demographic development
- Growing middle class due to above average economic development and rapid growth of population in Emerging Economies, especially in Asia
- Different degrees of development of airline business models
- Regional geographical setup
- …



Different Development in World Regions – Effects

- Rapid growth of air traffic in emerging markets (growing middle class)
- Slow growth in industrial countries due to saturated market
- Fast growth of low cost carriers in some regions
- New aircraft deliveries either for further growth or substitution of older aircraft
- Aircraft size might depend on geographical setup

Traffic growth in world regions



Source: Airbus GMF 2024

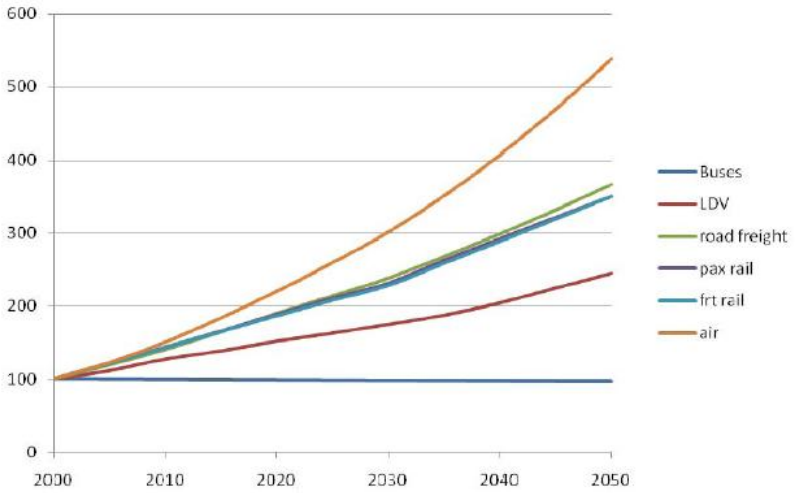
2. Introduction to Modern Civil Air Transport

2.3 Future Challenges in Aviation



Development of the Different Means of Transport

Global Traffic Development
(Passenger or Ton-km), Index (2000=100)



Source: ITF Transport Outlook 2010



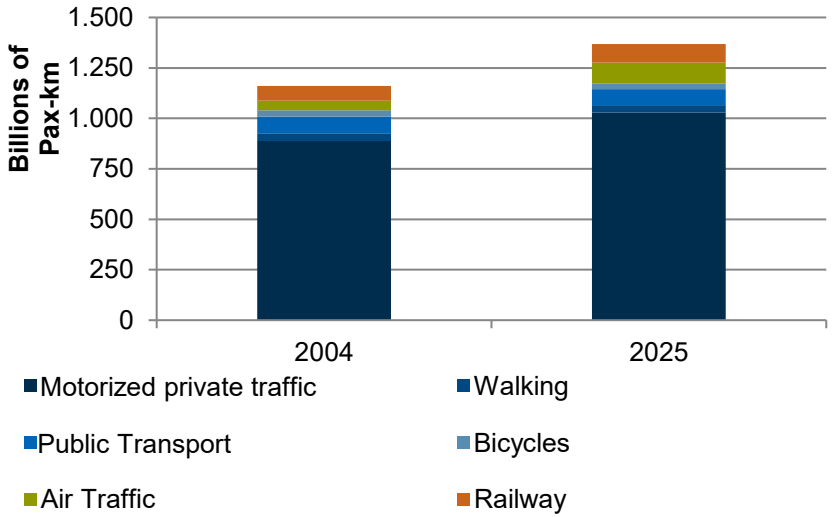




Aviation experiences the strongest growth wrt. pax- and ton-km.

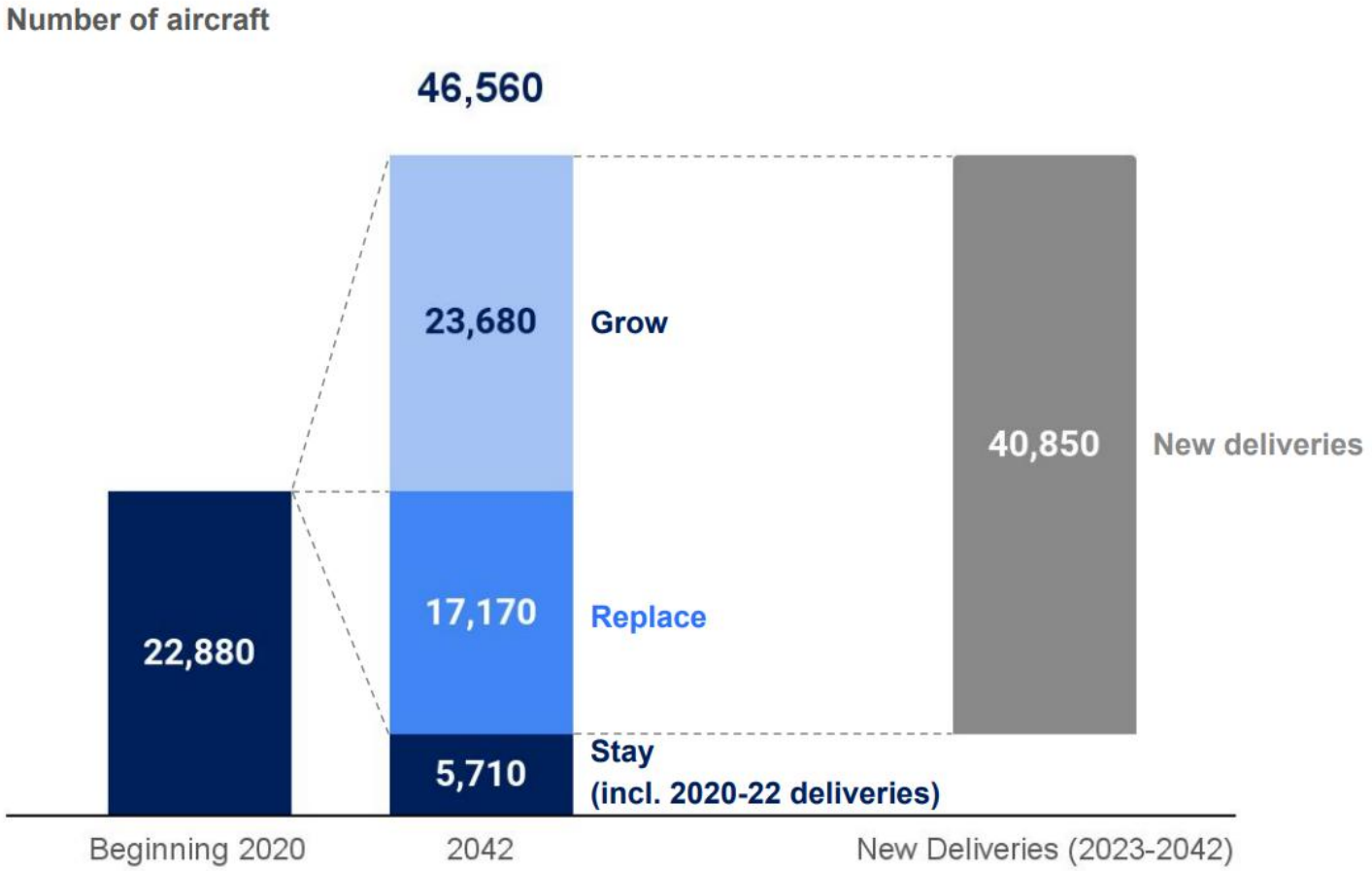
Little share in overall transport capacity.

Transport capacity in Germany



Source: Verkehr in Zahlen 2010/2011

Growth



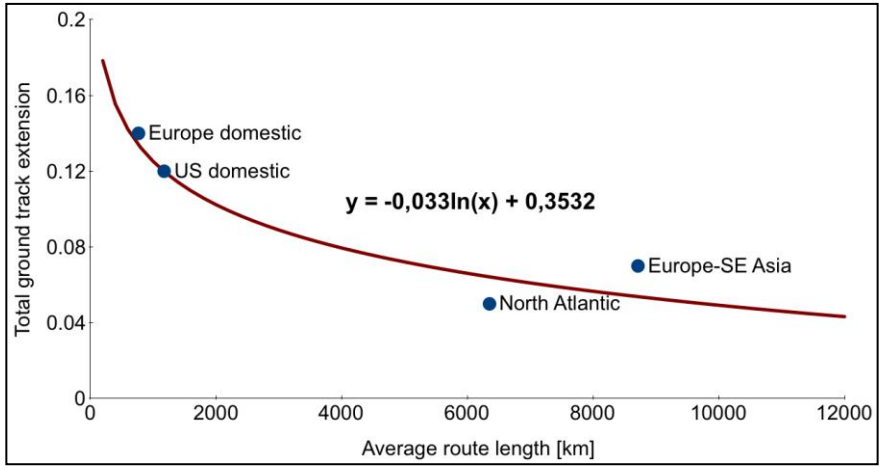
Source: Airbus GMF 2023

Capacity Constraints

In the air



Statistical function of **ground track extension** due to ATC restrictions (holdings, etc.)



On the ground

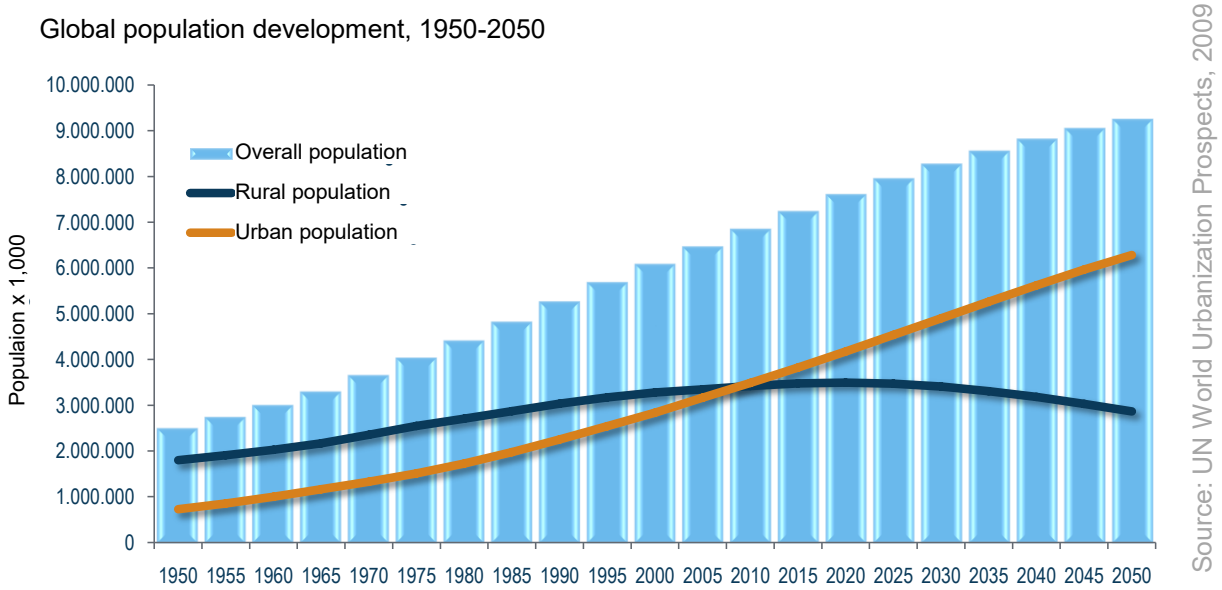


Top 5 international airports with **lowest flights on-time performance**

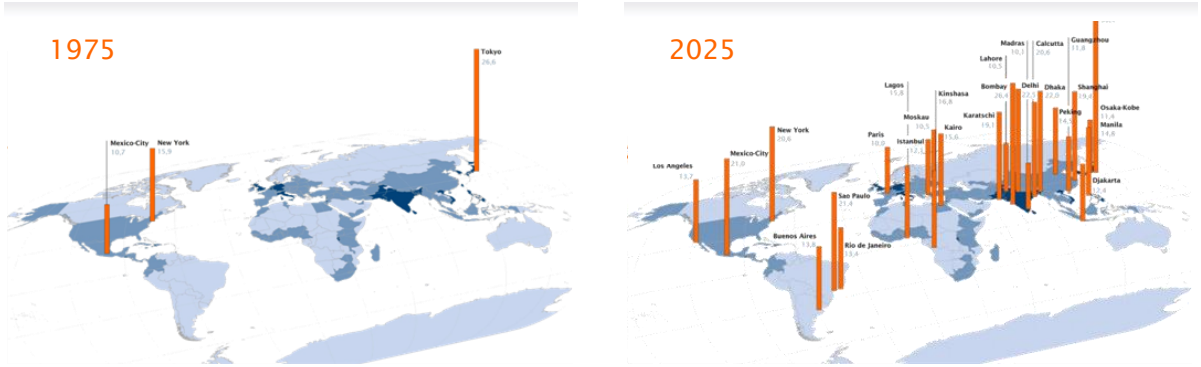
Airport Name (IATA Code)	Fraction of flights on time	Fraction of very late flights (30 – 44 min)	Fraction of excessively late flights (> 45 min)
Beijing (PEK)	38.25%	14.28%	12.86%
Shanghai (PVG)	39.20%	13.10%	16.27%
Istanbul (IST)	42.38%	11.79%	5.47%
Paris (CDG)	68.23%	6.27%	5.91%
Rome (FCO)	69.80%	5.93%	6.55%

Source: Reynolds (2009), FlightStats

Urbanization



Source: UN World Urbanization Prospects, 2009



Aviation:

Airport expansion is difficult

(new *green-field* projects are located far away from the cities)

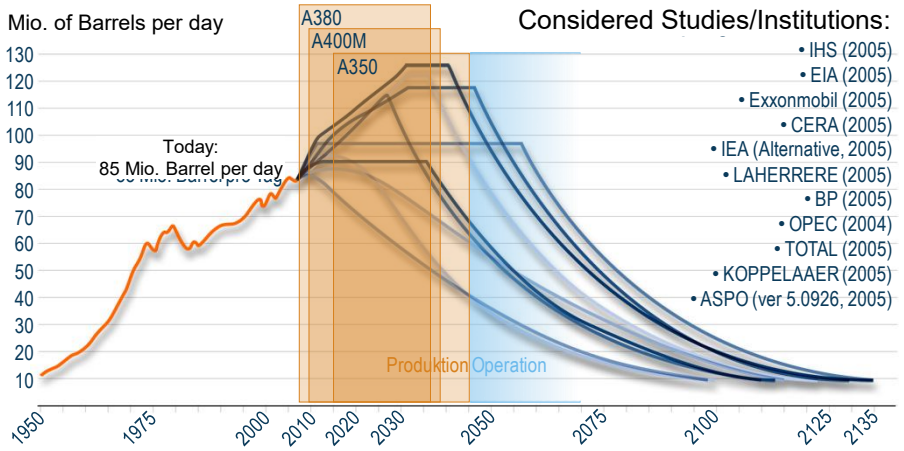
> Connection to the city centers restricts capacity



Shortage of Natural Resources – Oil

Peak Oil

- Precise point in time still unclear
- Current and upcoming aircraft types and technologies:
  → entirely dependent on oil



Alternative Aviation Fuels

- Biomass-to-Liquid (BTL), Hydro-treated Vegetable Oils (HVO)
- Target conflict: Food vs. Fuel

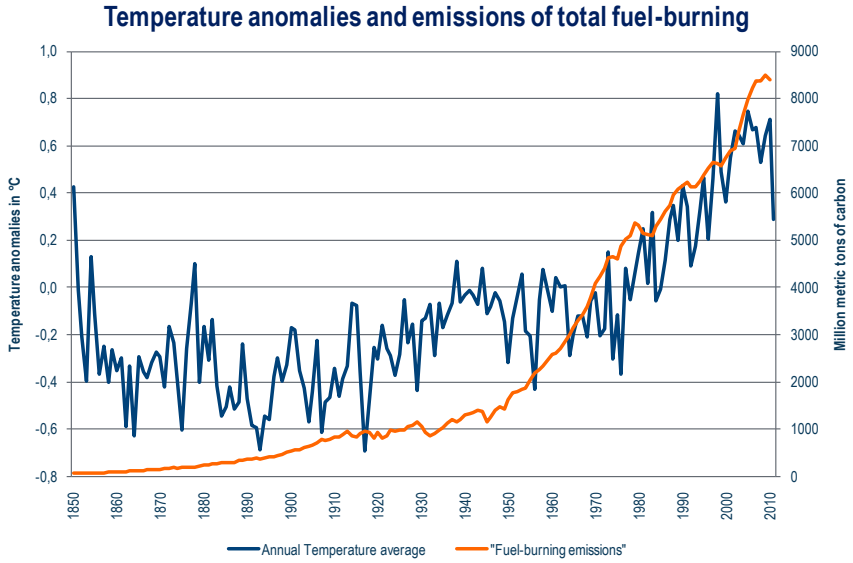






Images: Airbus

Climate Change



Source: *CDIAC* http://cdiac.ornl.gov/ftp/ndp030/global.1751_2007.ems;
IPCC http://www.cru.uea.ac.uk/cru/data/temperature/crutem3gl.txt





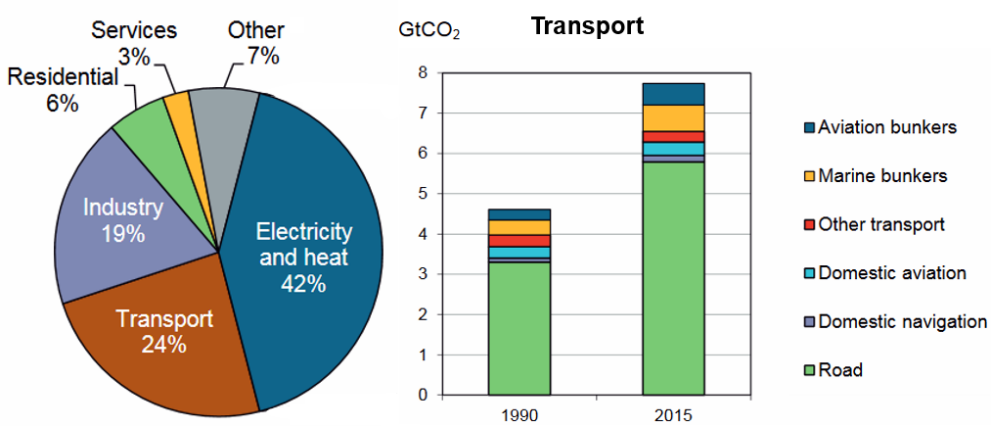
Aviation is committed to carbon-neutral growth from 2020.
Until 2050, aviation aims at cutting its carbon footprint by half relative to 2005.





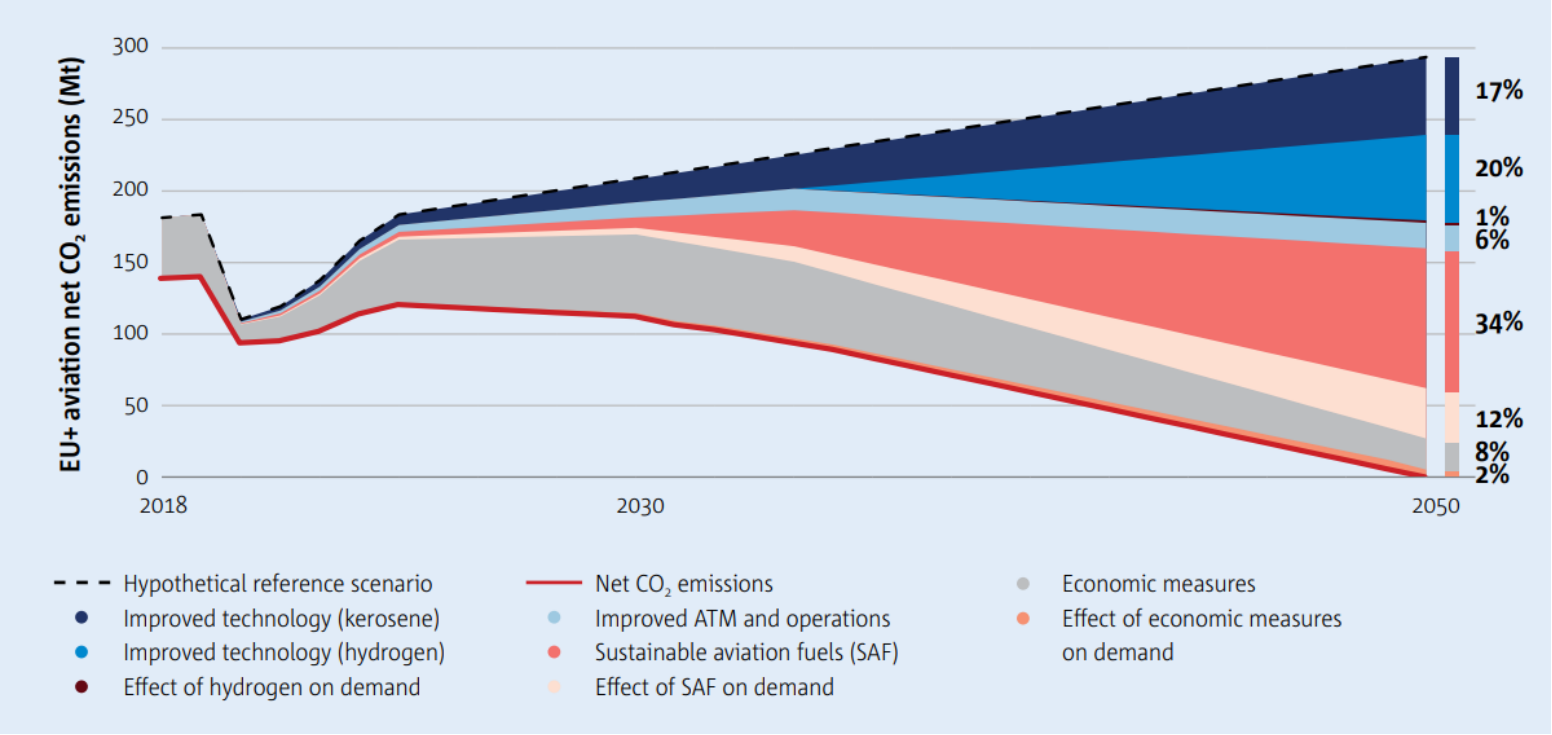
Images © fotolia

Global CO_2 emissions 2015 in %



Source: IEA

Emission-related Goals until 2050



These goals cannot be met solely by increasing operational efficiency.

- ✓ Alternative aviation fuels need to be introduced.
- ✓ Technological innovations are required on long term.

Source: EASA 2022

2. Introduction to Modern Civil Air Transport

2.4 Major Players



Major Players of the Civil Air Transport System

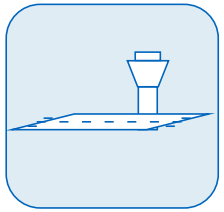

Maintenance

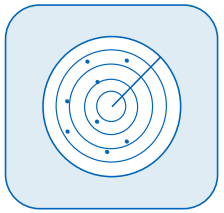

Original Equipment Manufacturers

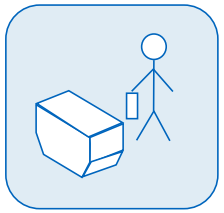

Authorities & Organizations


Operators


Aircraft Operations


Airports


Air Traffic Control


Passengers & Cargo


Environment

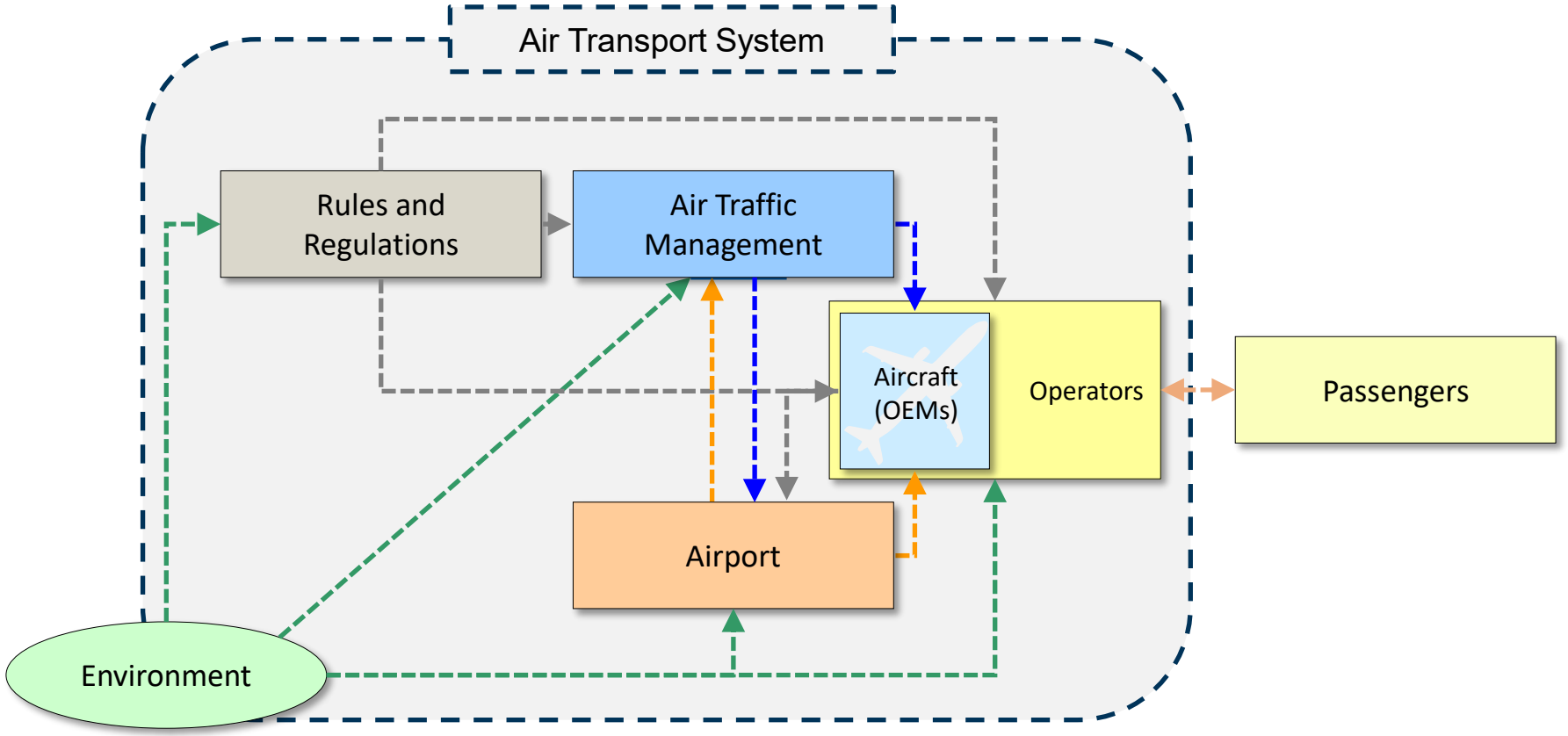
Interactions of the different Players in Aviation



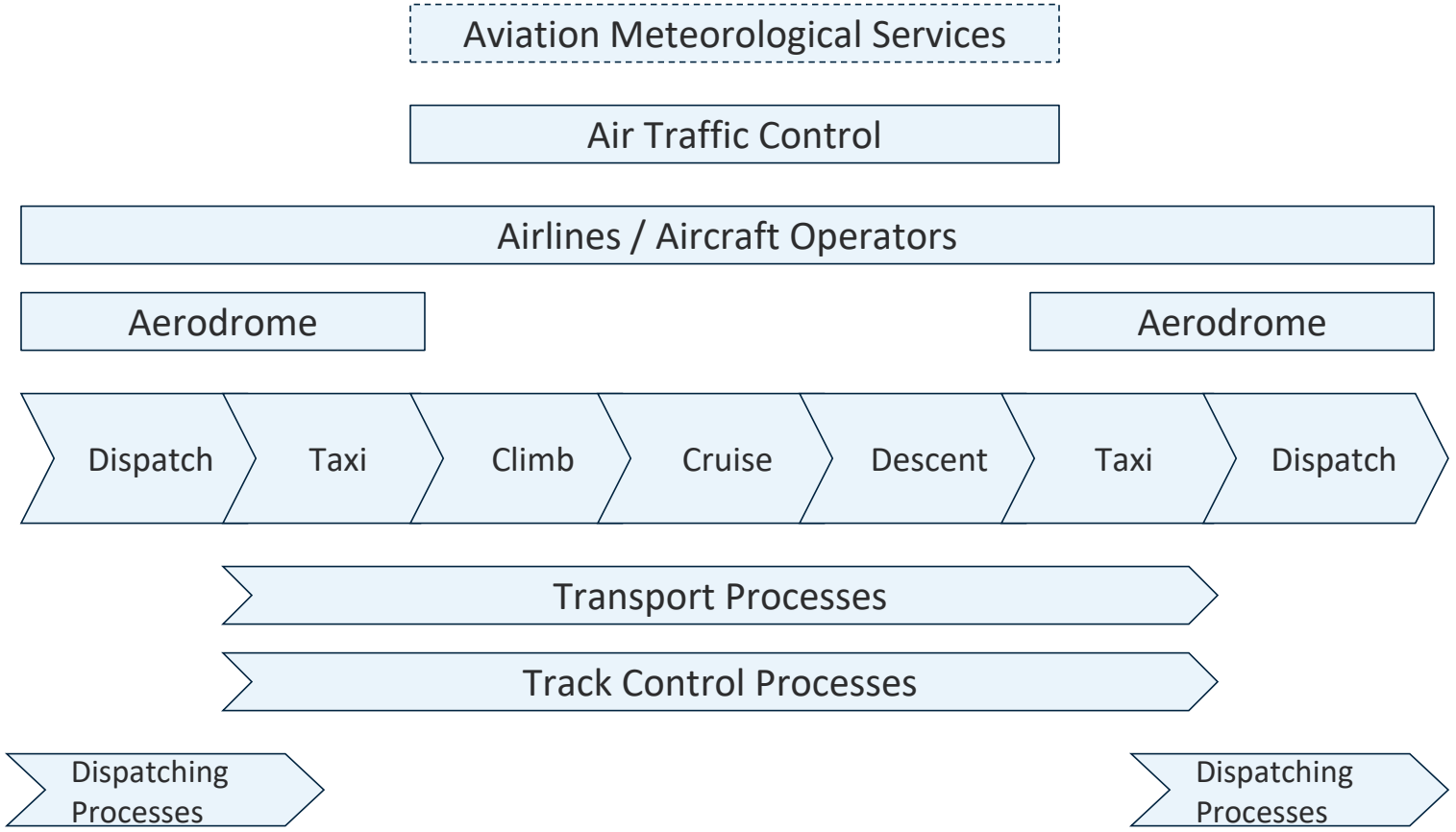
Planning, Developmet, Testing, and Operation of Air Vehicle Equipment,
Aerodromes, and Infrastructure for Air Traffic Management

Source: Mensen: Planung, Anlage und Betrieb von Flugplätzen

Interrelations in the Civil Air Transport System



Processes in the Civil Air Transport System



Source: Mensen: Planung, Anlage und Betrieb von Flugplätzen

Organizations in the Civil Air Transport System

	Singapore	*Asia*	*Europe*	*World*
General Transport Regulations	Ministry of Transport	--	European Commission Mobility and Transport Brussels	International Civil Aviation Organization (ICAO) Montreal
Certification, Airworthiness, Air Crews	Civil Aviation Authority of Singapore (CAAS)	--	European Aviation Safety Agency (EASA) Cologne	Federal Aviation Administration (FAA) USA
Air Traffic Management	CAAS	--	Eurocontrol Maastricht	Future Air Navigation Systems (FANS)
Airline	Singapore Airlines, Scoot, Silk Air, Jetstar Asia Airways	Association of Asia Pacific Airlines (AAPA) Kuala Lumpur	Association of European Airlines (AEA) Brussels	International Air Transport Association (IATA) Montreal
Airport	CAAS Changi Airport (SIN) Seletar Airport (XSP)	Airports Council International – Asia-Pacific (ACI-Asia-Pacific) Hong Kong	Airports Council International – Europe (ACI-Europe) Brussels	Airports Council International (ACI) Geneva
OEM	--	Comac	Airbus Group	Boeing, Airbus, Embraer, etc.

Global Organizations

- ICAO was established in 1947 in response to the Chicago Convention
- World headquarters: Montreal (Canada), eight regional offices
- Specialized agency of the UN, 191 member states
- Most important functions:
  - Development, approval, and updating of international technical standards and recommended practices for airports and air traffic control
  - Publication of regulatory guidelines and of economic and environmental policy statements regarding international air transport

- Trade association of most of the international airlines in the world, founded in Havana (Cuba) in April 1945.
- Headquarters in Geneva and Montreal
- IATA membership: 287 airlines in 124 countries, flights by IATA members represent 82% of total traffic (Available Seat Kilometers)
- IATA operates as a trade association offering various technical, legal, and financial services
- Still operating as a “tariff coordination” organization

Source: ICAO, Belobaba (2009)

European Agencies

- Drafting aviation safety legislation and providing technical advice to the European Commission and to the Member States
- Inspections and training
- Airworthiness and environmental type-certification and design approvals of aircraft and aeronautical products globally
- Coordination and analysis of safety programs

EASA
European Aviation Safety Agency

EUROCONTROL

- Harmonize and integrate air navigation services in Europe
- Discussion and implementation of Single European Sky
- Platform for civil-military coordination
- Support day-to-day operations of the European ATM
- Coordinate response to crisis situations in Europe
- Provide public service functions (e.g. route charges)

Source: EASA, Eurocontrol

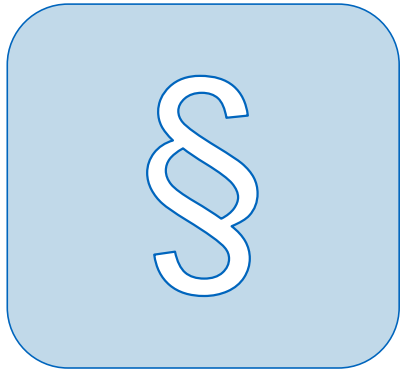
US Government Agencies

- **US Department of Transportation (US DOT):** responsible for domestic and international economic policy and regulation
- **US Department of State:** provides support in the area of international aviation policy and regulation
- **US Department of Justice:** responsible for issues involving antitrust legislation, including the prosecution of alleged violators
- **Federal Aviation Administration (FAA):** six lines of activity: regulation and certification, research and acquisitions, airports, administration, commercial space transportation, and air traffic services
- **National Transportation Safety Board (NTSB):** charged with investigating every civil aviation accident in the USA, as well as all other major transportation accidents, and with making recommendations on improving transportation safety
- **Transportation Security Administration (TSA):** Protect the Nation's transportation systems to ensure freedom of movement for people and commerce

Source: Belobaba (2009), TSA

2. Introduction to Modern Civil Air Transport

2.5 Rules and Regulations



Convention on International Civil Aviation

- **Chicago, 1944**: 54 states attended the International Convention on Civil Aviation
- Introduction of the “Chicago Convention,” an international treaty that laid the foundation for today’s global air transportation system
- The Convention confirmed that aviation was of potentially enormous global importance.
- A decision had to be made between a “liberal” and a “protectionist” regulatory environment for international services (> much controversy ever since)
- The convention decided to create a framework for regulating air transport services **on a bilateral basis**
- Result: bilateral **Air Service Agreements (ASAs)** = instrument for initiating and modifying international transportation services and for regulating these services
- 1946, “Bermuda I” Agreement: established the ground rules for US-UK air services
- Bilateral ASAs are still prevailing today, with multilateral ASAs becoming increasingly important in recent years
- ICAO was founded in order to ensure international commonality in airport and air traffic control facilities, as well as equipment and procedures

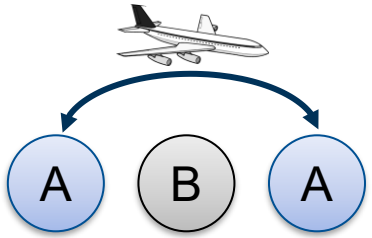
Source: Belobaba (2009)

Freedoms of the Air… (1)

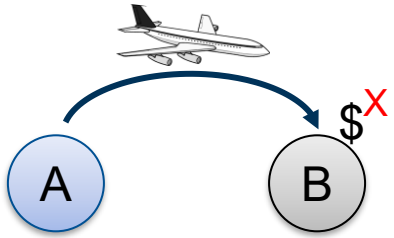
…refer to the rights that an airline of any state may enjoy with respect to another state(s).

Airline is registered in state A

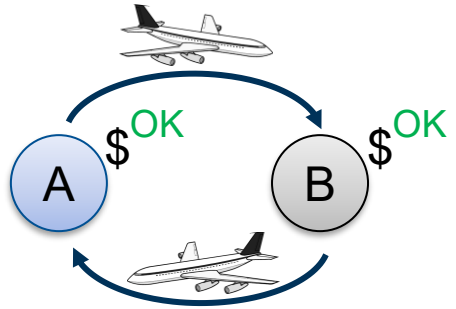
1st Freedom
Fly over a state B without landing.



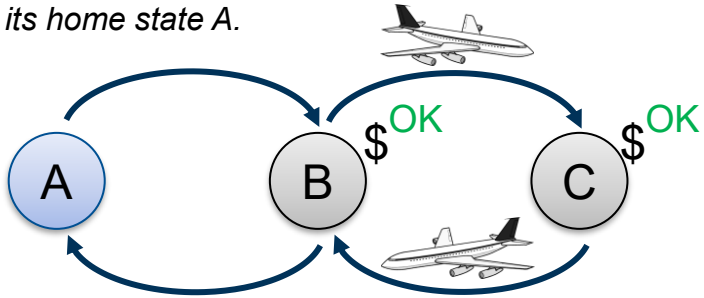
2nd Freedom
Land in state B without picking up or setting down any revenue traffic (e.g., refueling, maintenance).



3rd and 4th Freedom
Enplane revenue traffic from its home state A for transport to an airport in state B, and vice versa.



5th Freedom
Enplane revenue traffic at an airport of state B for transport to a third state C, and vice versa, *as part of the continuation of a service originating or terminating in its home state A.*



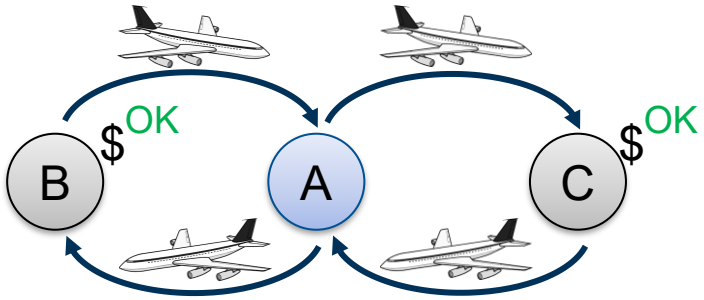
Source: Belobaba (2009)

Freedoms of the Air… (2)

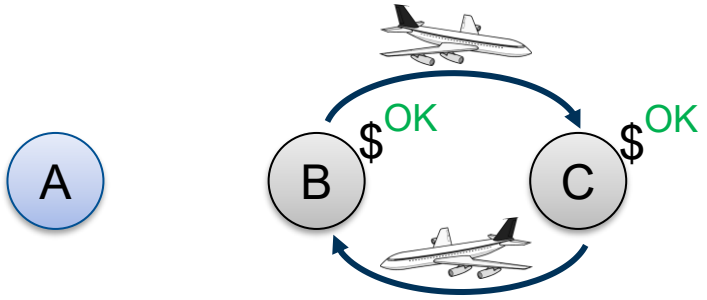
…refer to the rights that an airline of any state may enjoy with respect to another state(s).

Airline is registered in state A

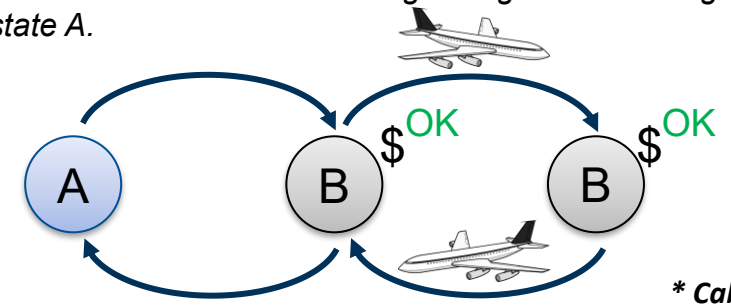
6th Freedom
Use separate sets of 3rd and 4th Freedom rights with two other states B and C in order to transport revenue traffic between these two other states *using an airport in A as a connection point.*



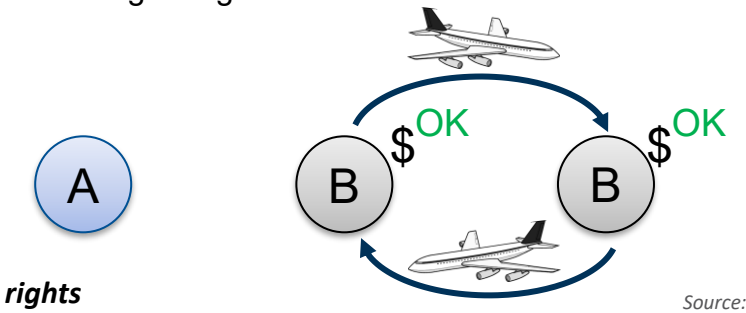
7th Freedom
Transport revenue traffic between a pair of airports in states B and C on a service that operates entirely outside state A.



8th Freedom*
Carry *domestic* revenue traffic between two points within state B *on a service originating or terminating in state A.*



9th Freedom*
The same as the 8th Freedom, but with no requirement that the flight originates or terminates in home state A.



**** Cabotage rights***

Source: Belobaba (2009)

Aviation Legislation – Overview

Licensing Regulations

Development

- Aircraft (Licensing, Airworthiness, Maintenance), Flight Performance: FAA / EASA
- Avionics: DO 178B, DO 254
- Components: ISO TC 20 Aircraft and Space Vehicles

Operations

- EASA Part-OPS
- ICAO
- FAA (e.g., ETOPS)
- IATA

Licensing of Companies (Development, Manufacturing, Maintenance)

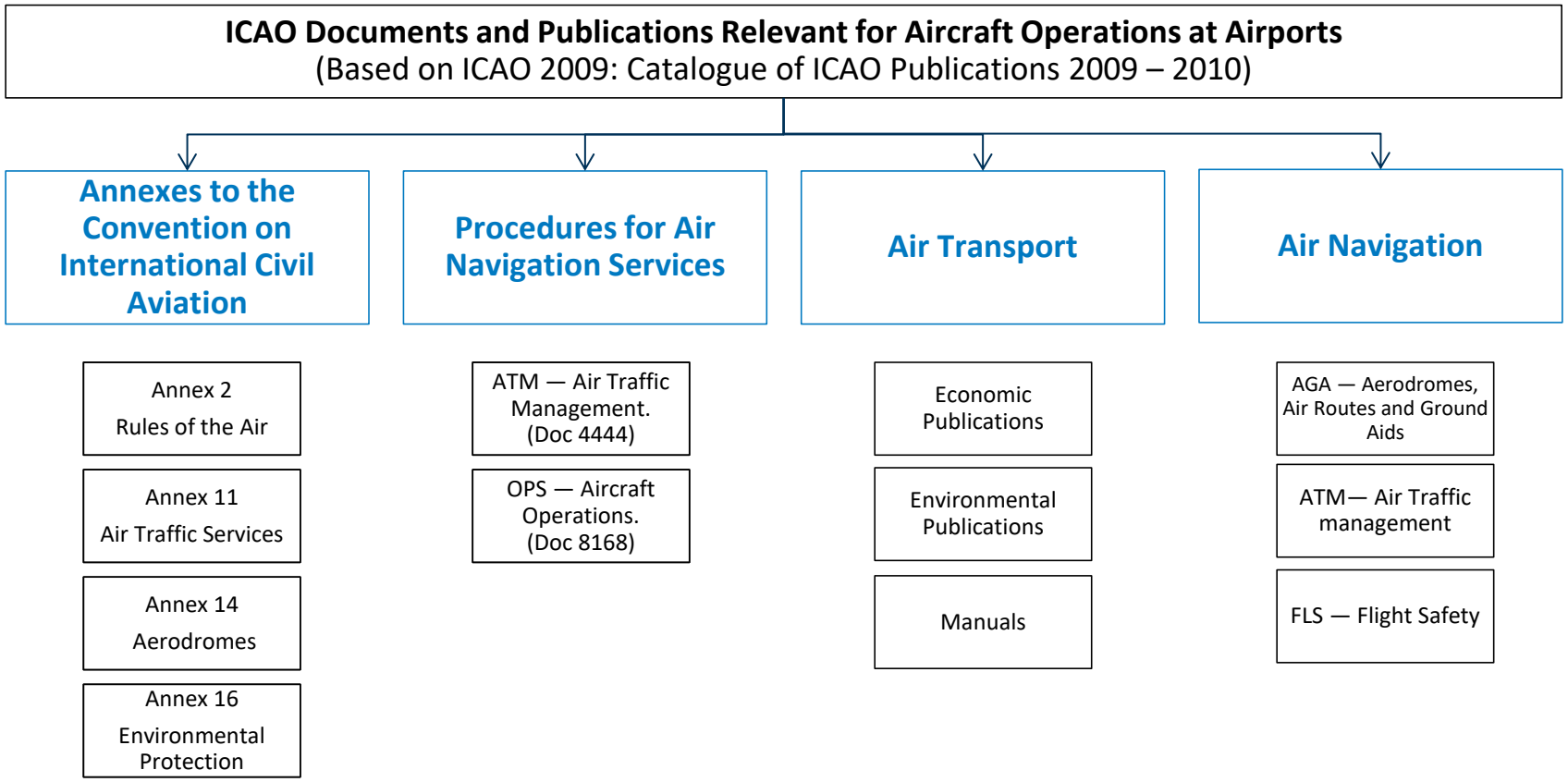
- EASA

ICAO Annexes to the Chicago Convention

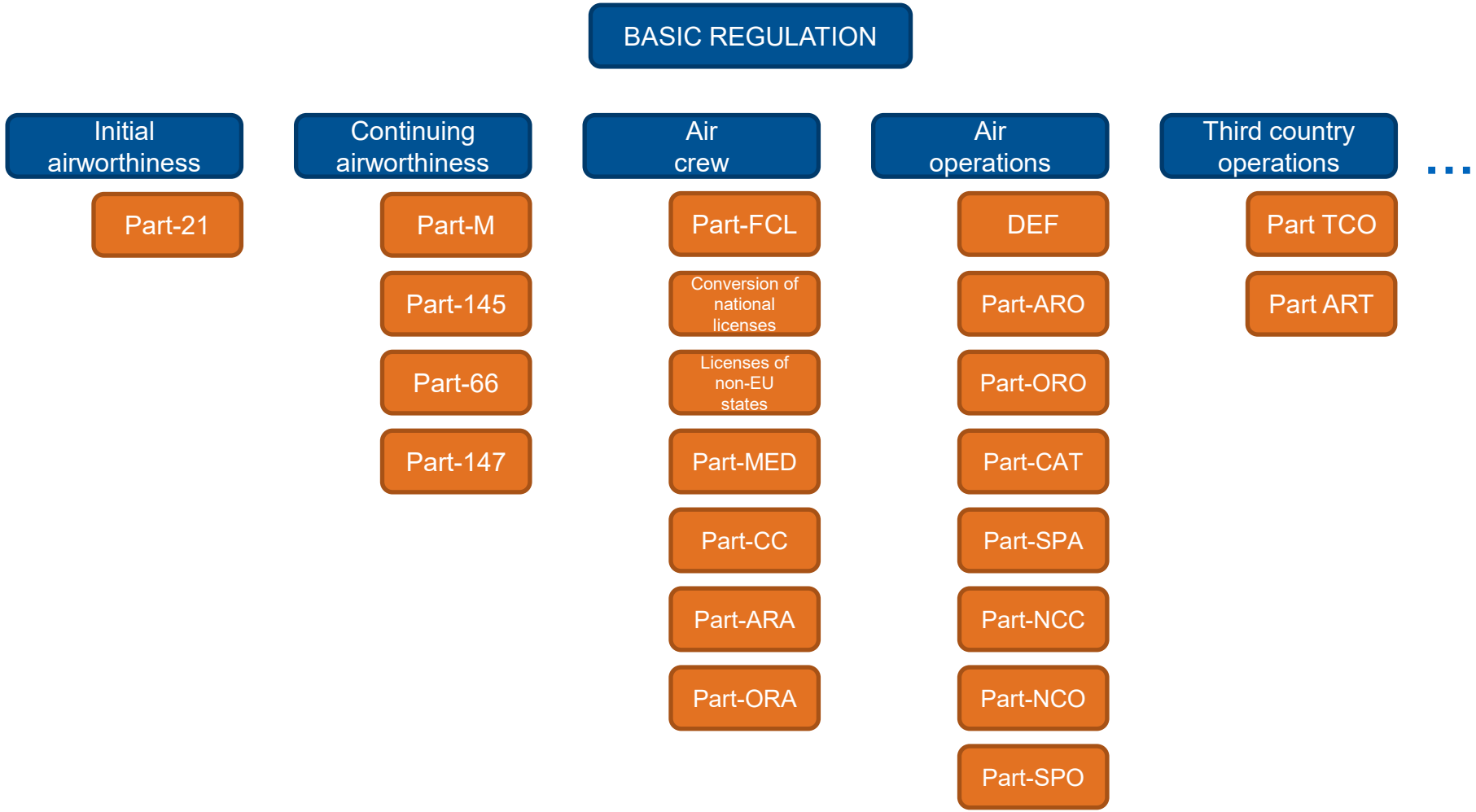
Annex 1	Personnel Licensing
Annex 2	Rules of the Air
Annex 3	Meteorological Service for International Air Navigation
Annex 4	Aeronautical Charts
Annex 5	Units of Measurement to be Used in Air and Ground Operations
Annex 6	Operation of Aircraft
Annex 7	Aircraft Nationality and Registration Marks
Annex 8	Airworthiness of Aircraft
Annex 9	Facilitation
Annex 10	Aeronautical Telecommunications
Annex 11	Air Traffic Services
Annex 12	Search and Rescue
Annex 13	Aircraft Accident and Incident Investigation
Annex 14	Aerodromes
Annex 15	Aeronautical Information Services
Annex 16	Environmental Protection
Annex 17	Security: Safeguarding International Civil Aviation Against Acts of Unlawful Interference
Annex 18	The Safe Transport of Dangerous Goods by Air

Source: ICAO

ICAO – Air Operations Regulations

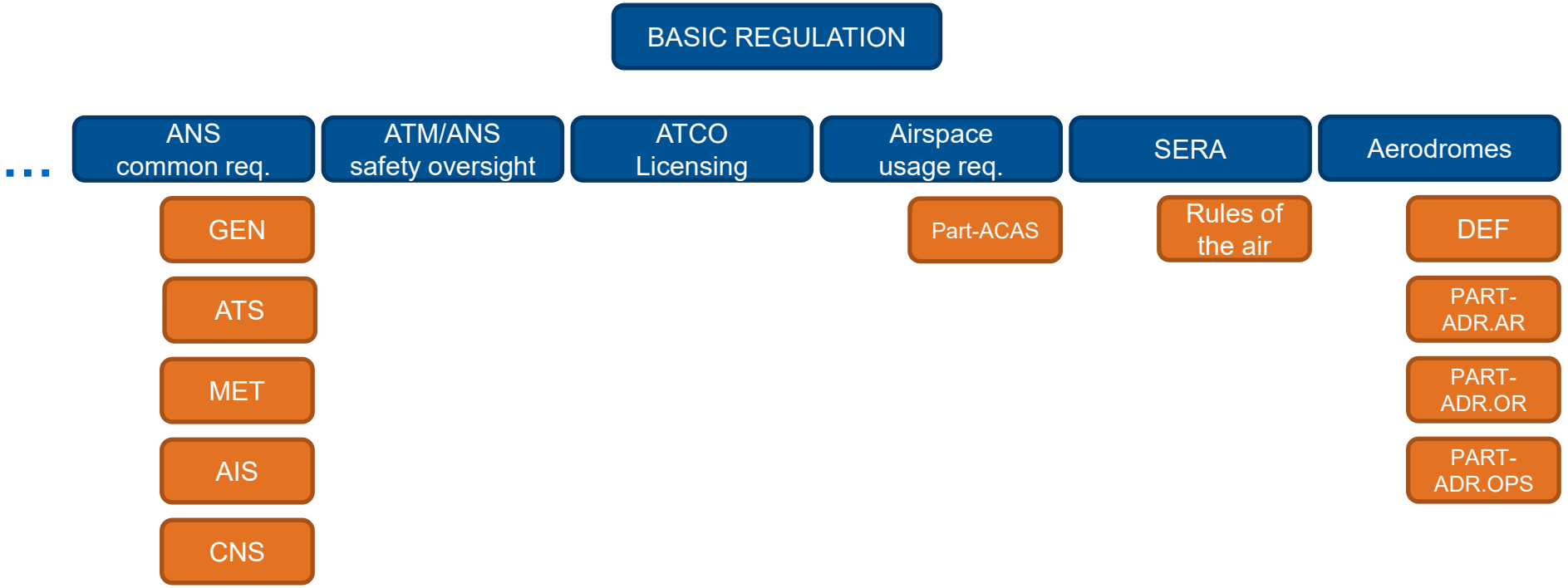


EASA Regulations Structure (1)



Source: EASA

EASA Regulations Structure (2)



ANS… Air Navigation Services

ATM… Air Traffic Management

ATCO… Air Traffic Controller

SERA… Services and Procedures in Air Navigation

Source: EASA

EASA Certification Specifications

CS-22	Sailplanes and Powered Sailplanes
CS-23	Normal, Utility, Aerobatic and Commuter Aeroplanes
CS-25	Large Aeroplanes
CS-27	Small Rotorcraft
CS-29	Large Rotorcraft
CS-31HB	Hot Air Balloons
CS-34	Aircraft Engine Emissions and Fuel Venting
CS-36	Aircraft Noise
CS-APU	Auxiliary Power Units
CS-AWO	All Weather Operations
CS-E	Engines
CS-ETSO	European Technical Standard Orders
CS-Definitions	Definitions and Abbreviations
CS-P	Propellers
CS-VLA	Very Light Aeroplanes
CS-VLR	Very Light Rotorcraft
AMC-20	General Acceptable Means of Compliance for Airworthiness of Products, Parts and Appliances

CS-25 Large Aeroplanes

BOOK 1 – AIRWORTHINESS CODE
SUBPART A – GENERAL
SUBPART B – FLIGHT
SUBPART C – STRUCTURE
SUBPART D – DESIGN AND CONSTRUCTION
SUBPART E – POWERPLANT
SUBPART F – EQUIPMENT
SUBPART G – OPERATING LIMITATIONS AND INFORMATION
SUBPART J – GAS TURBINE AUXILIARY POWER UNIT INSTALLATION
APPENDIX H – INSTRUCTIONS FOR CONTINUED AIRWORTHINESS
APPENDIX I – AUTOMATIC TAKEOFF THRUST CONTROL SYSTEM (ATTCS)
APPENDIX J – EMERGENCY DEMONSTRATION

BOOK 2 – ACCEPTABLE MEANS OF COMPLIANCE(AMC):
INTRODUCTION
SUBPART B – FLIGHT
SUBPART C – STRUCTURE
SUBPART D – DESIGN AND CONSTRUCTION
SUBPART E – POWERPLANT
SUBPART F – EQUIPMENT
SUBPART G – OPERATING LIMITATIONS AND INFORMATION
SUBPART J – GAS TURBINE AUXILIARY POWER UNIT INSTALLATION
AMC APPENDIX F

2. Introduction to Modern Civil Air Transport

2.6 Safety



Definitions

“Accident” (ICAO)

An occurrence associated with the operation of an aircraft which, in the case of a manned aircraft, takes place between the time any person boards the aircraft with the intention of flight until such time as all such persons have disembarked, or in the case of an unmanned aircraft, takes place between the time the aircraft is ready to move with the purpose of flight until such time as it comes to rest at the end of the flight and the primary propulsion system is shut down, in which:

a) **A person is fatally or seriously injured** as a result of:
- Being in the aircraft, or
- Direct contact with any part of the aircraft, including parts which have become detached from the aircraft, or
- Direct exposure to jet blast,

except when the injuries are from natural causes, self-inflicted or inflicted by other persons, or when the injuries are to stowaways hiding outside the areas normally available to the passengers and crew; or

b) **The aircraft sustains damage or structural failure** which:
- Adversely affects the structural strength, performance or flight characteristics of the aircraft, and
- Would normally require major repair or replacement of the affected component,

except for engine failure or damage, when the damage is limited to a single engine, (including its cowlings or accessories), to propellers, wing tips, antennas, probes, vanes, tires, brakes, wheels, fairings, panels, landing gear doors, windscreens, the aircraft skin (such as small dents or puncture holes), or for minor damages to main rotor blades, tail rotor blades, landing gear, and those resulting from hail or bird strike (including holes in the radome); or

c) **The aircraft is missing or is completely inaccessible.**

Definitions

“Serious Injury” (ICAO)

An injury which is sustained by a person in an accident and which:

a) **Requires hospitalization** for more than 48 hours, commencing within seven days from the date the injury was received

b) Results in a **fracture of any bone** (except simple fractures of fingers, toes or nose)

c) Involves **lacerations** which cause severe hemorrhage, nerve, muscle, or tendon damage

d) Involves **injury to any internal organ**

e) Involves **second or third degree burns**, or **any burns affecting more than 5 percent of the body surface**

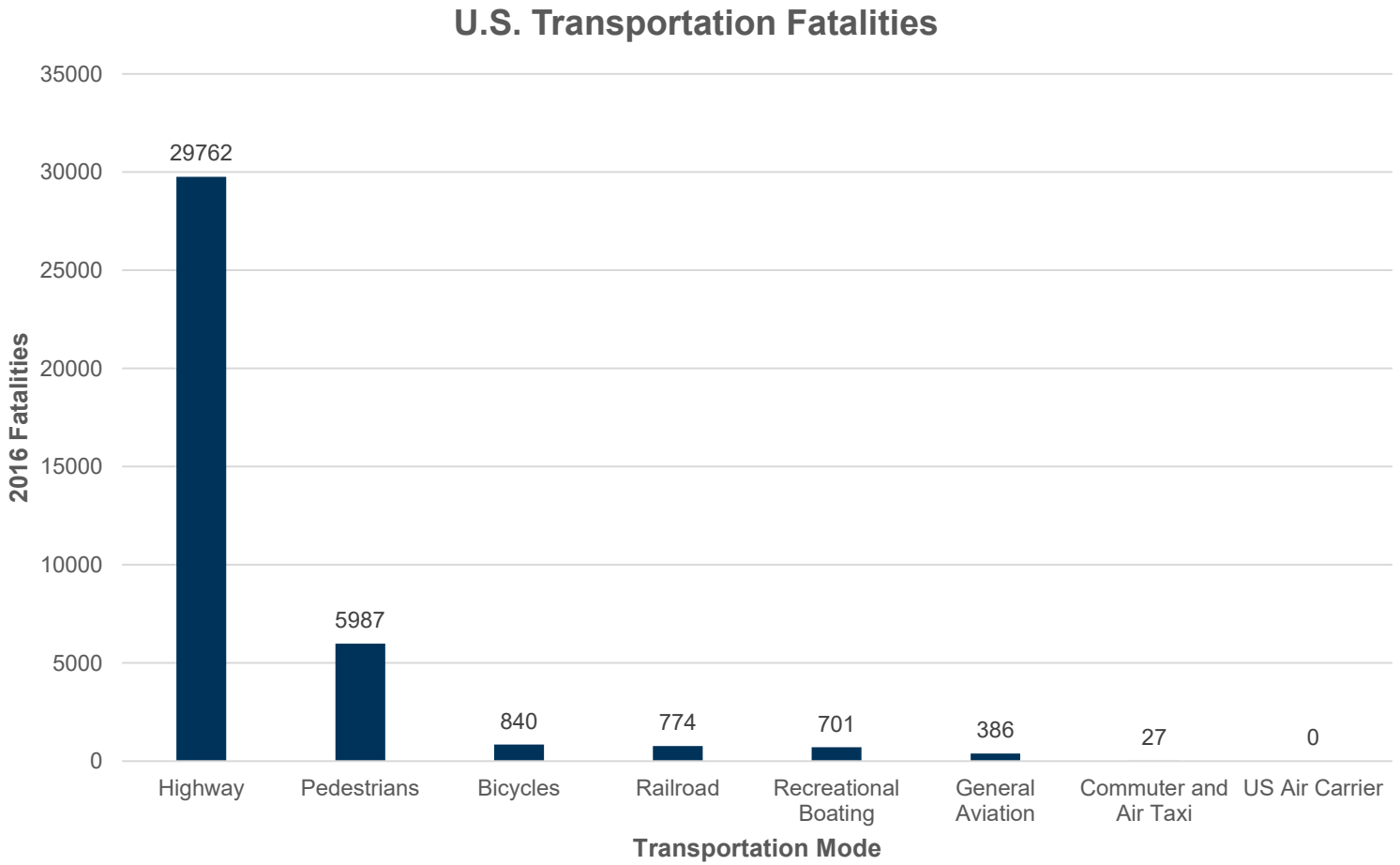
f) Involves **verified exposure to infectious substances or injurious radiation**.

“Substantial Damage” (NTSB)

Substantial damage means damage or failure **which adversely affects the structural strength, performance, or flight characteristics of the aircraft**, and which would normally require major repair or replacement of the affected component.

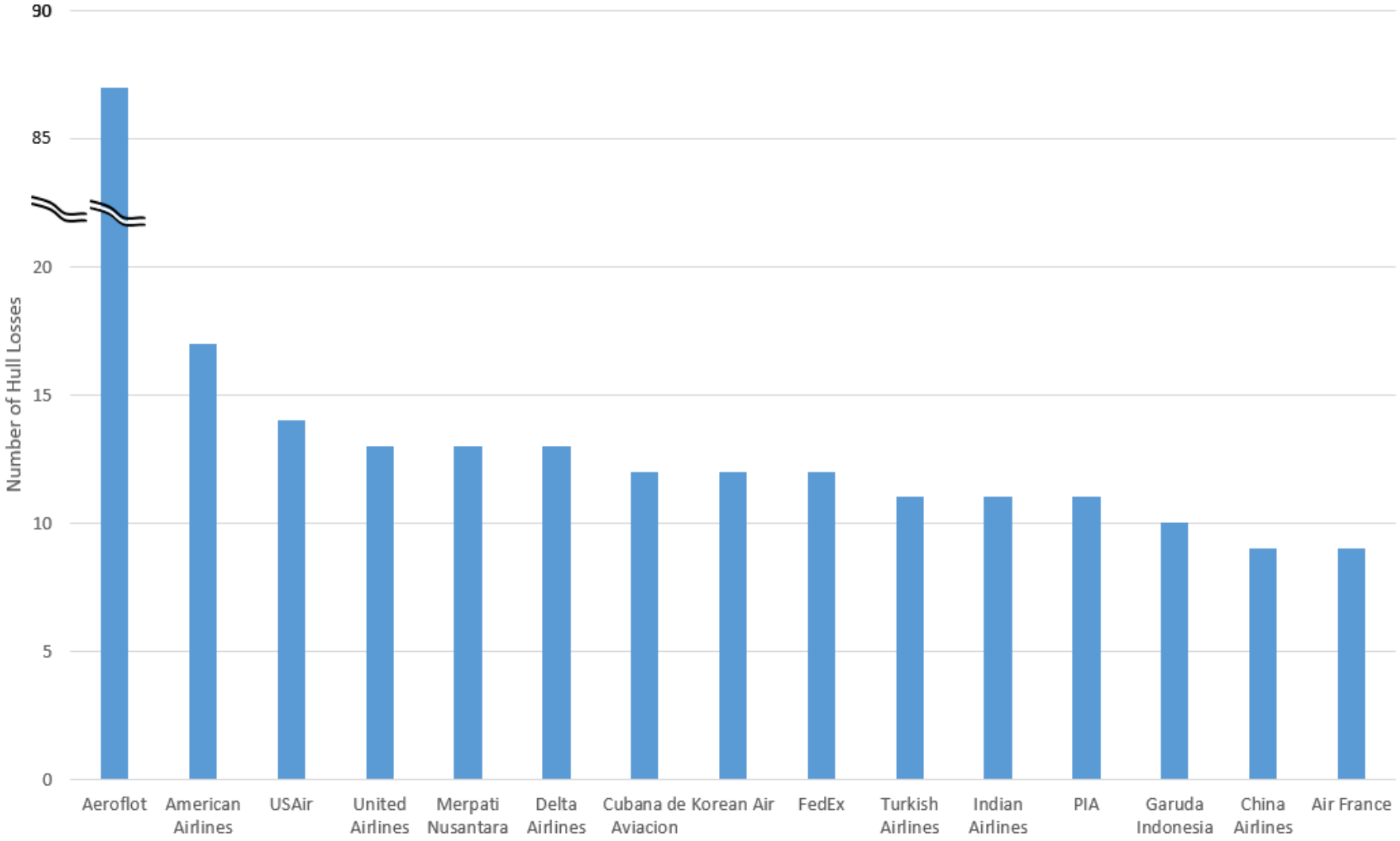
Engine failure or damage limited to an engine if only one engine fails or is damaged, bent fairings or cowling, dented skin, small puncture holes in the skin or fabric, ground damage to rotor or propeller blades, and damage to landing gear, wheels, tires, flaps, engine accessories, brakes, or wingtips are not considered ‘substantial damage’ for the purpose of this part.

“Airline Travel is Extraordinarily Safe.”



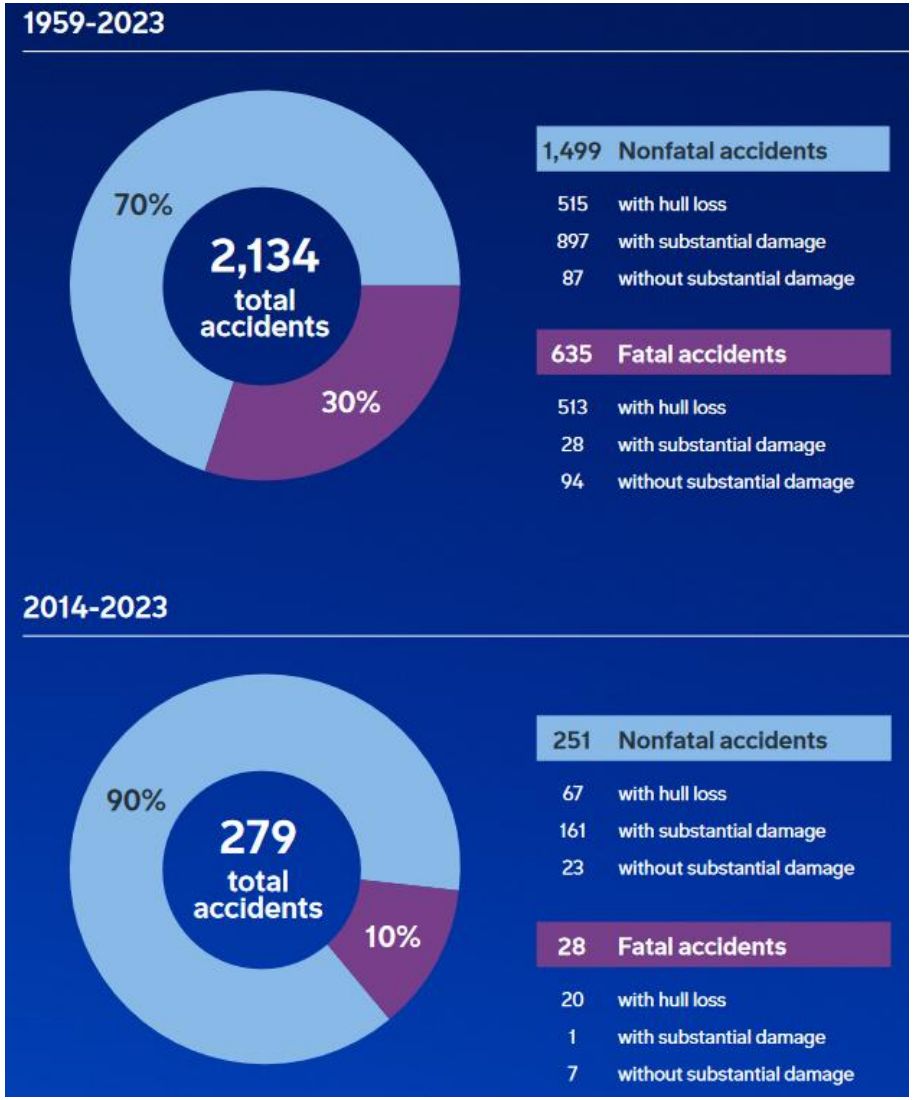
Source: Bureau of Transportation Statistics

Airlines with Highest Amount of Aircraft Losses 1983-2018



Source: Aviation Safety Network

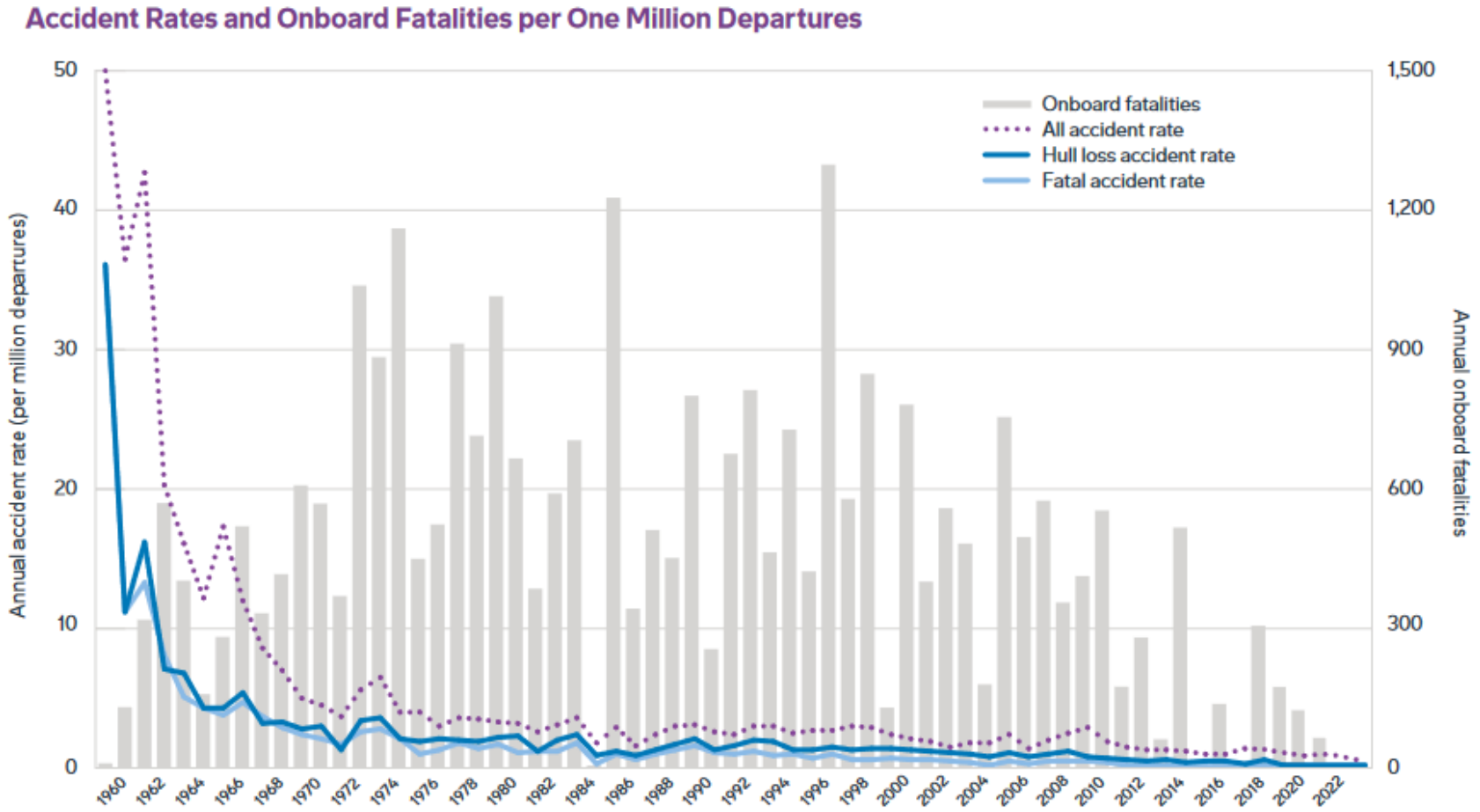
Accident Summary by Injury and Damage



Source: Boeing – Statistical Summary of Commercial Jet Airplane Accidents 1959-2023

Accident Rates and Onboard Fatalities by Year

Worldwide commercial jet fleet: 1959 – 2023



Source: Boeing – Statistical Summary of Commercial Jet Airplane Accidents 1959-2023

Accident Numbers by Type of Operation

Fatal and hull loss accidents, worldwide commercial jet fleet

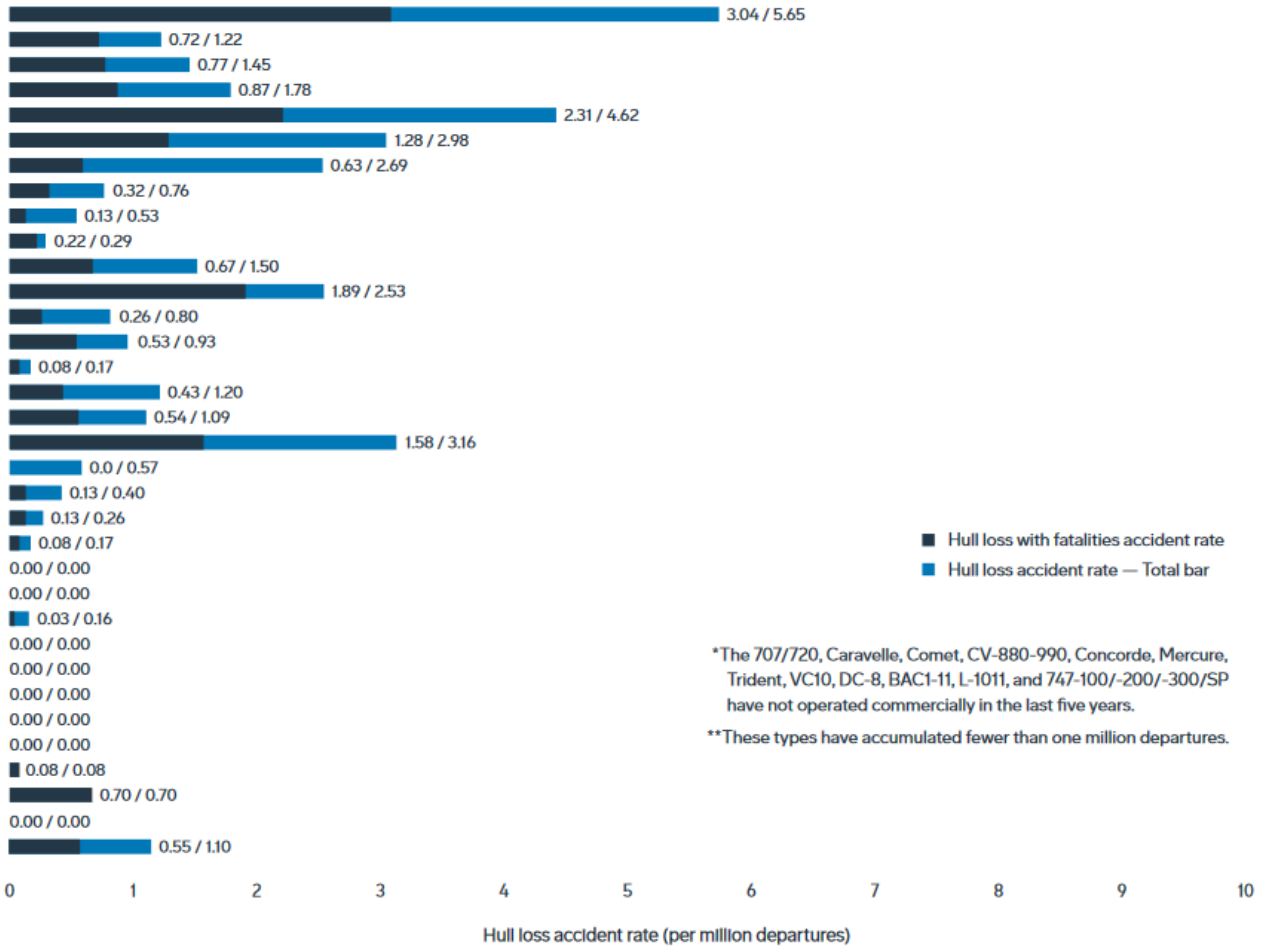
Type of Operation	All Accidents		Fatal Accidents		Onboard Fatalities (External Fatalities)*		Hull Loss Accidents	
	1959-2023	2014-2023	1959-2023	2014-2023	1959-2023	2014-2023	1959-2023	2014-2023
Passenger	1,704	230	510	24	29,643 (810)	1,066 (19)	757	63
– Scheduled	1,580	226	463	23	25,446 (806)	995 (19)	686	62
– Charter	124	4	47	1	4,197 (4)	71 (0)	71	1
Cargo	308	47	83	4	285 (385)	12 (43)	197	22
Maintenance test ferry, positioning, training and demonstration	122	2	42	0	190 (66)	0 (0)	74	2
Totals	**2,134**	**279**	**635**	**28**	**30,118 (1, 261)**	**1,078 (62)**	**1,028**	**87**

Source: Boeing – Statistical Summary of Commercial Jet Airplane Accidents 1959-2023

Accident Rates by Airplane Type

Hull Loss Accidents | Worldwide Commercial Jet Fleet | 1959-2023

Sorted by Year of Introduction	Hull Loss	Hull Loss With Fatalities	
*No Longer in Service	392	211	3.04 / 5.65
727	94	55	0.72 / 1.22
DC-9	91	48	0.77 / 1.45
737-100/-200	104	51	0.87 / 1.78
F-28	42	21	2.31 / 4.62
DC-10/MD-10	28	12	1.28 / 2.98
A300	17	4	0.63 / 2.69
MD-80/-90	36	15	0.32 / 0.76
767	12	3	0.13 / 0.53
757	8	6	0.22 / 0.29
BAe 146, RJ-70/-85/-100	18	8	0.67 / 1.50
A310	12	9	1.89 / 2.53
737-300/-400/-500	62	20	0.26 / 0.80
A300-600	7	4	0.53 / 0.93
A320/321/319/318	29	13	0.08 / 0.17
F-100/F-70	14	5	0.43 / 1.20
747-400	10	5	0.54 / 1.09
MD-11	10	5	1.58 / 3.16
A340	2	0	0.0 / 0.57
A330	6	2	0.13 / 0.40
777	4	2	0.13 / 0.26
737-600/-700/-800/-900	22	10	0.08 / 0.17
717	0	0	0.00 / 0.00
CRJ-700/-900/-1000	0	0	0.00 / 0.00
EMB-170/-175/-190	5	1	0.03 / 0.16
**A380	0	0	0.00 / 0.00
787	0	0	0.00 / 0.00
**747-8	0	0	0.00 / 0.00
A350	0	0	0.00 / 0.00
C-Series/A220	0	0	0.00 / 0.00
A320/321/319 NEO	1	1	0.08 / 0.08
737 MAX	2	2	0.70 / 0.70
A330 NEO	0	0	0.00 / 0.00
Total	**1,028**	**513**	0.55 / 1.10



*The 707/720, Caravelle, Comet, CV-880-990, Concorde, Mercure, Trident, VC10, DC-8, BAC1-11, L-1011, and 747-100/-200/-300/SP have not operated commercially in the last five years.

**These types have accumulated fewer than one million departures.

Source: Boeing – Statistical Summary of Commercial Jet Airplane Accidents 1959-2023

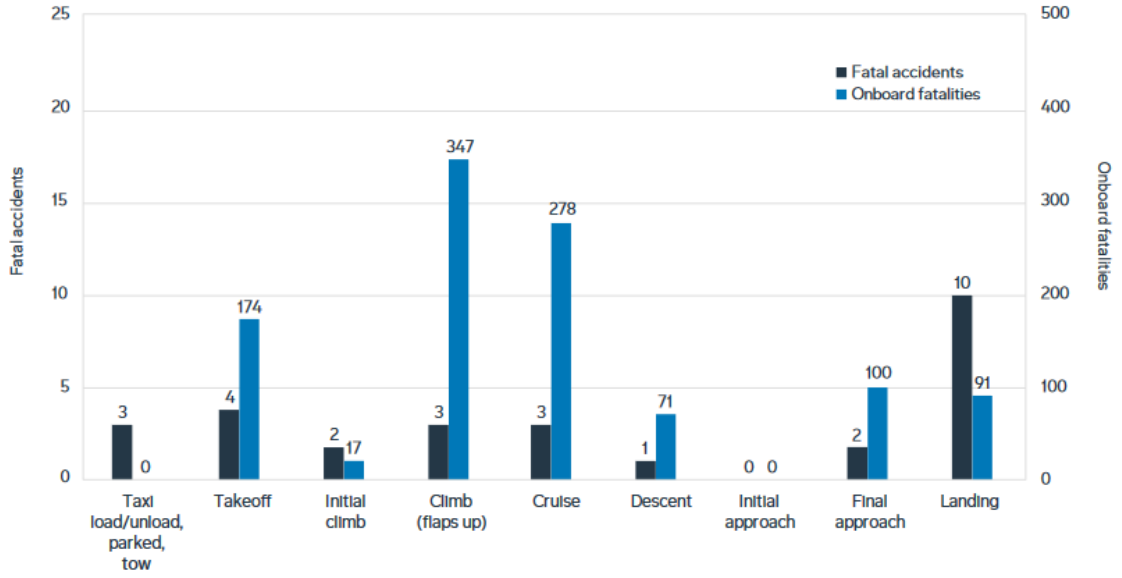
Fatal Accidents and Onboard Fatalities by Phase of Flight

Percentage of Fatal Accidents and Onboard Fatalities | 2014-2023

	Taxi, load/ unload, parked, tow	21% Takeoff	Initial climb	Climb (flaps up)	Cruise	Descent	Initial approach	43% Final approach	Landing
Fatal accidents	11%	14%	7%	11%	11%	4%	0%	7%	36%
Onboard fatalities	0%	16%	2%	32%	26%	7%	0%	9%	8%
		18%						17%	
Exposure (percentage of flight time estimated for a 1.5-hour flight)		1%	1%	14%	57%	*Initial approach fix* 11%	*Final approach fix* 12%	3%	1%

Note: Percentages may not sum to 100% because of numerical rounding.

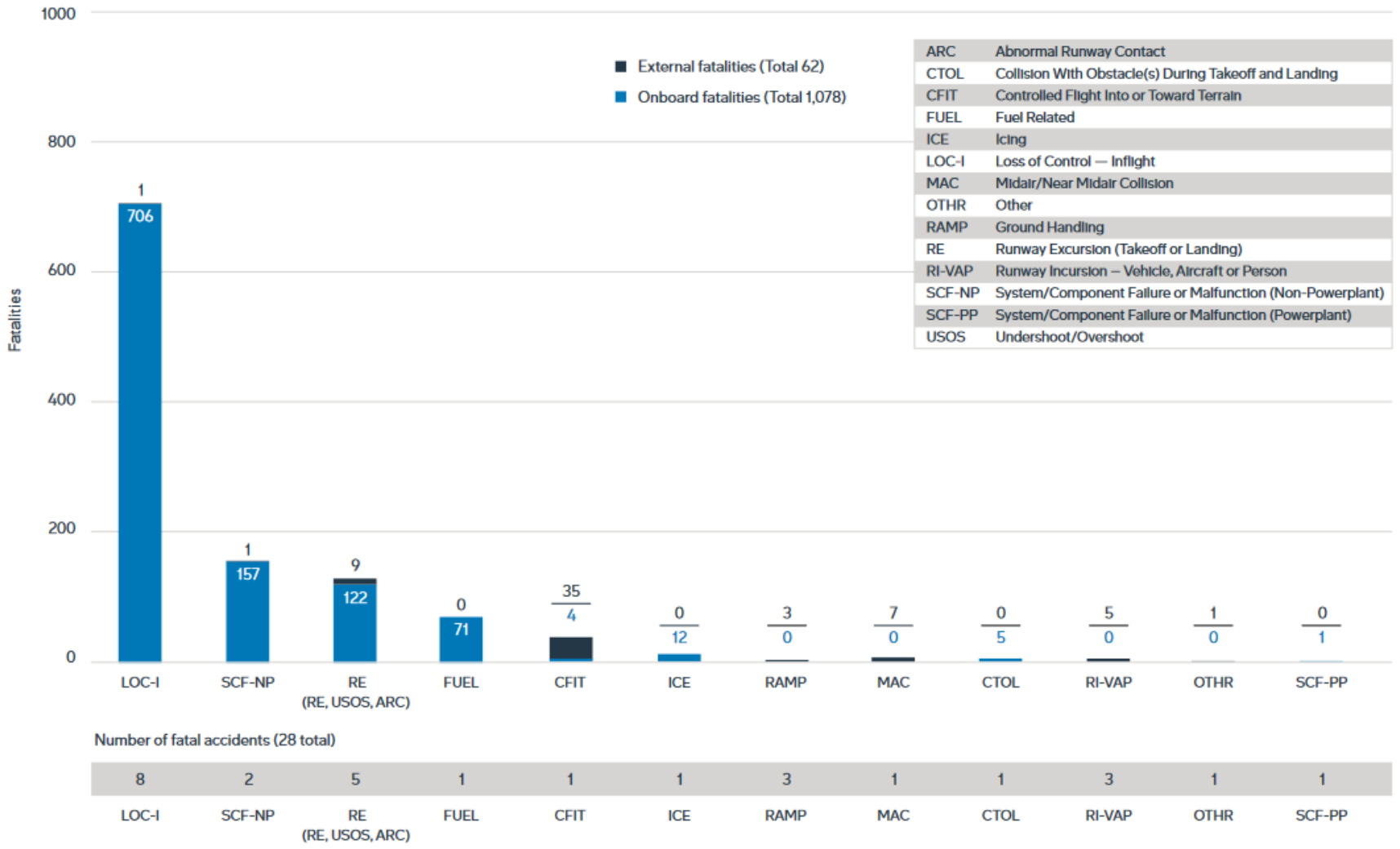
Distribution of Fatal Accidents and Onboard Fatalities | 2014-2023



Source: Boeing – Statistical Summary of Commercial Jet Airplane Accidents 1959-2023

Fatalities by CAST/ICAO Common Taxonomy Team Aviation Occurrence Categories

Fatal Accidents | Worldwide Commercial Jet Fleet | 2014-2023

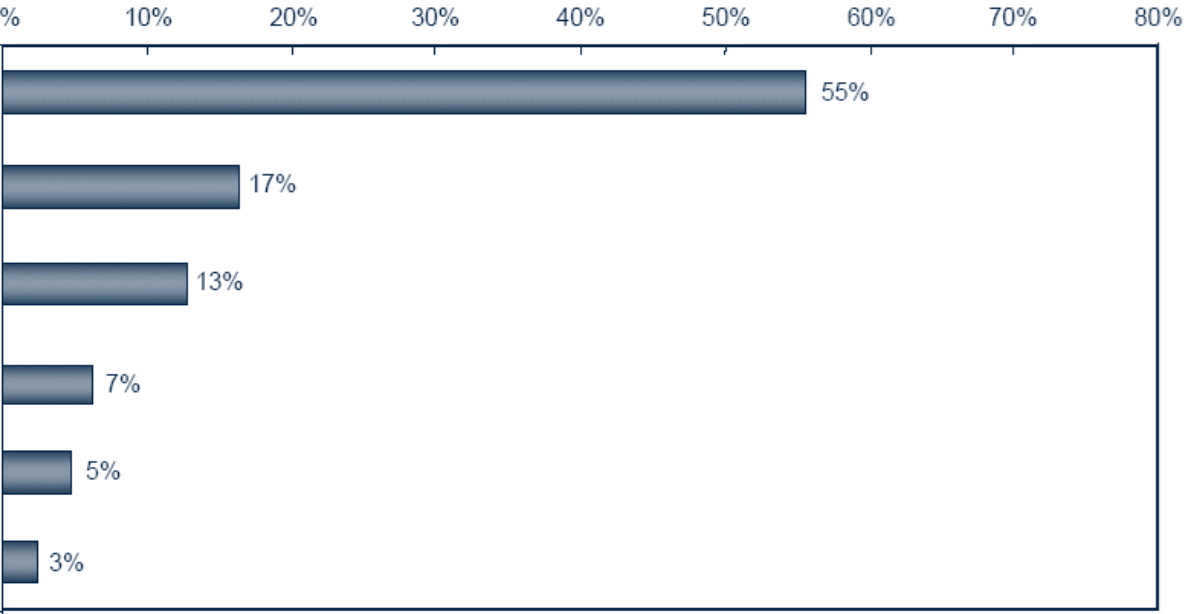


ARC	Abnormal Runway Contact
CTOL	Collision With Obstacle(s) During Takeoff and Landing
CFIT	Controlled Flight Into or Toward Terrain
FUEL	Fuel Related
ICE	Icing
LOC-I	Loss of Control – Inflight
MAC	Midair/Near Midair Collision
OTHR	Other
RAMP	Ground Handling
RE	Runway Excursion (Takeoff or Landing)
RI-VAP	Runway Incursion – Vehicle, Aircraft or Person
SCF-NP	System/Component Failure or Malfunction (Non-Powerplant)
SCF-PP	System/Component Failure or Malfunction (Powerplant)
USOS	Undershoot/Overshoot

Source: Boeing – Statistical Summary of Commercial Jet Airplane Accidents 1959-2023

Primary Reasons for Total Losses

Worldwide commercial jet fleet: 1994 – 2005



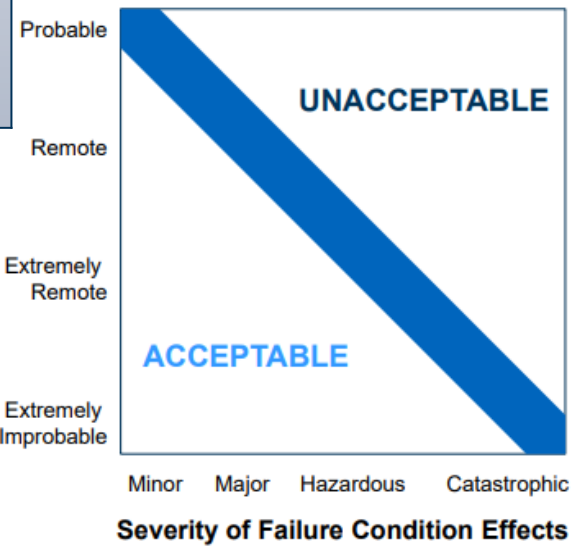
Total with known causes	134
Unknown or awaiting reports	49
Total	183

*As determined by the investigating authority, percent of accidents with known causes.

Source: Statistical Summary of Commercial Jet Airplane Accidents, Boeing

Relation between Probability and Impact

Probability of occurrence per flight										
	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}	10^{-9}	10^{-10}
Probability	Frequent		Moderately probable		Less probable		Improbable		Very improbable	
Impact	Minor				Major		Risky		Catastrophic	
Description	Will probably happen frequently during the life span of one aircraft unit		Will probably happen less frequently during operation, but can happen several times during the life span of one aircraft unit		Will probably not happen during operation, but can happen several times during the life span of some aircraft units		Improbable to happen during the life span of one aircraft unit, but needs to be considered possible		Improbable in a way that it does not need to be considered possible	



Probability of Occurrence

Example

No. of civil aircraft in operation:	approx. 20,000 units
Average flight hours per aircraft (AC) p.a.:	approx. 3,000 FH / a / AC
➢ **Flight hours per year:**	**$6 \cdot 10^7$ FH / a**

From a statistical point of view, one highly improbable event with risky impact happens once a year at global level.

Expected life span of an aircraft:	approx. 20 a
➢ Expected flight hours during life span:	**$6 \cdot 10^4$ FH / AC**

From a statistical point of view, only one moderately or less probable system failure with at maximum a minor or major impact will happen during the entire life span of an aircraft.

Controlled Flight into Terrain (CFIT)

Until impacting the terrain, the aircraft is being under technical control, but not under navigational control.

CFIT is the cause for 50% of all fatal aviation accidents.

CFIT situations are not solely linked to an erroneous behavior of the pilots. Frequently, unclear information from “outside” is the beginning of CFIT accidents.

Results of an analysis of CFIT accidents:

- Air Traffic Control (ATC) was involved in 11 of 12 accidents.
- Only in 1 of 100 accidents, the pilots acted against the instructions from ATC.



Safety

Safety has to be thoroughly organized and institutionalized.

Hence, the following is needed:

- an independent supervisory authority
- Quality Management in aircraft operations
- Safety awareness at management level

The degree of success of the implementation of safety-related strategies and policies can be measured with statistical analyses.

An **Incident Reporting System** can help reduce the number of accidents.

In complex systems, the **interfaces** (e.g., human-machine interfaces HIDs) are the critical elements.

Incident Reporting System (IRS)

Key requirements

- An IRS has to ensure confidentiality.
- Reports have to be protected and should not be traceable to a person.
- Pilot reports must go unpunished.
- An IRS has to be operated and maintained by a neutral authority.
- Consequences have to be analyzed and evaluated carefully.
- The collection and analysis of data should be done without involving a regulative authority.
- Airlines should try and establish a “culture of safety.”
- The system has to remain flexible and should not cause an excessive level of bureaucracy.